CDE2501 | Liveable Cities

Week 3 | Studio 1 (Introduction)

Welcome to Liveable Cities

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3As Agenda

1. Assessment
2. Admin
3. Assist

1. Break/ Consult

ASSESSMENT

Assessment Breakdown

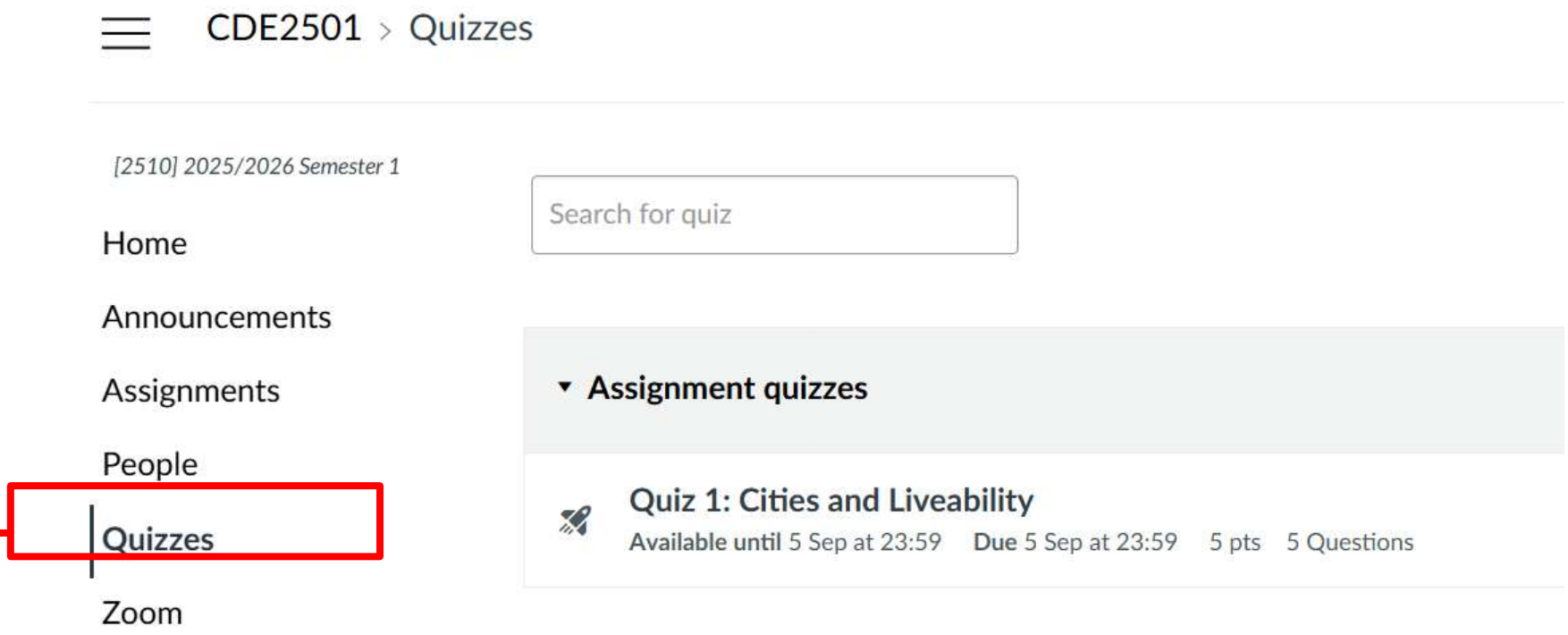
	Grading Component	Weightage	
Group	Presentation 1	15%	40%
	Final Presentation	25%	
Individual	Individual Presentation	10%	60%
	Individual Contribution & Engagement	15%	
	Peer Review	10%	
	Seminar-based Quizzes & Assignments	25%	

Seminar-based Individual Assignments & Quizzes

Quiz/Assignment	Date/Time that Quiz/Assignment will be open on Canvas	Topic of Quiz/Assignment
Quiz 1*	Fri 23 Jan (During Seminar 2)	Seminar 2
Quiz 2*	Fri 6 Feb (During Seminar 3)	Seminar 3
Quiz 3*	Fri 13 Mar (During Seminar 4)	Seminar 4
Quiz 4*	Fri 20 Mar (During Seminar 5)	Seminar 5
Individual Assignment 1	Fri 6 Feb – Fri 13 Feb (due 11:59pm)	Reflection on Seminars 1-3
Individual Assignment 2	Fri 20 Mar – Fri 27 Mar (due 11:59pm)	Reflection on Seminars 4-5

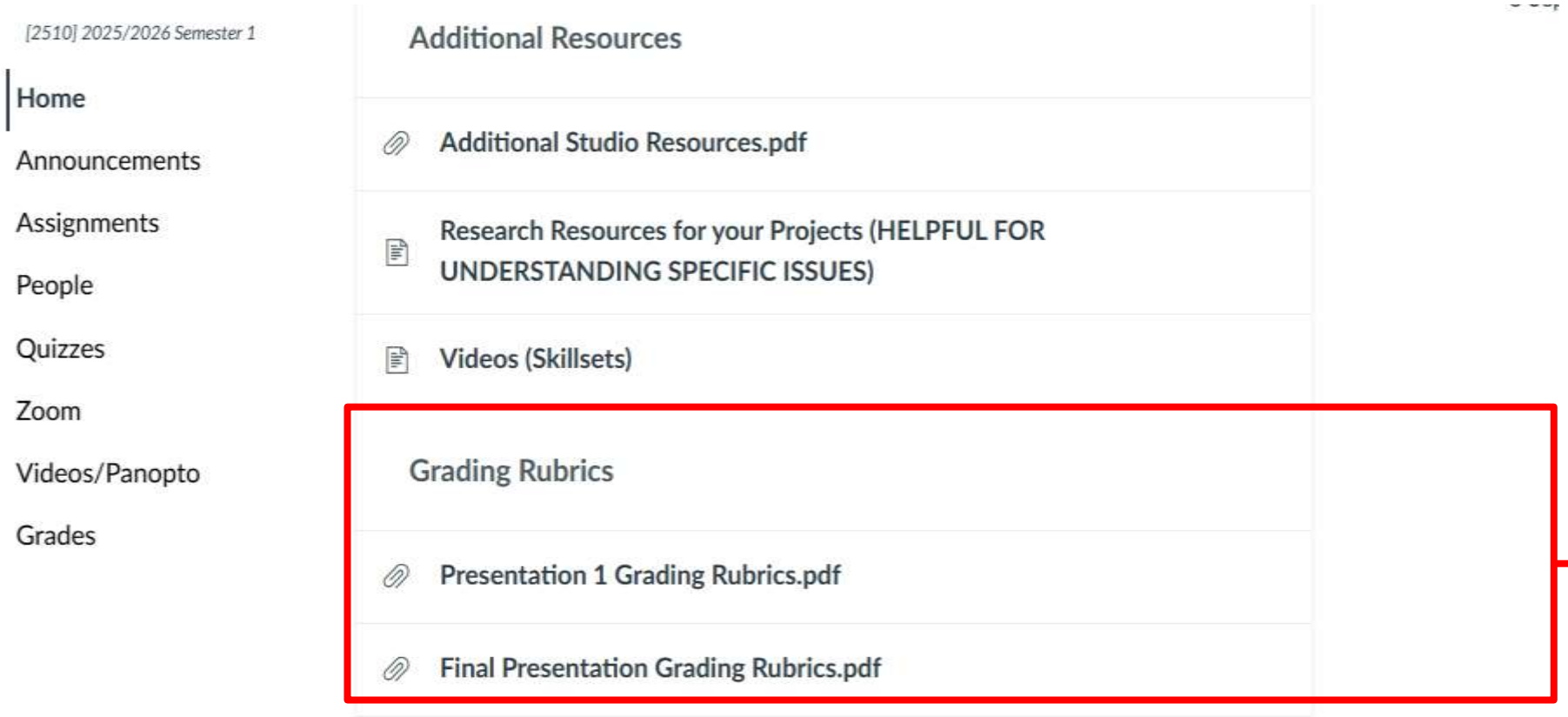
You can find seminar-based individual assignments here:

CANVAS sidebar > Quizzes



Studio Project (Group) Grade

- You will be graded based on the rubrics which can be found on Canvas



Rubrics for each presentation can also be found on Canvas, so you can know how you will be assessed

Individual Presentation – 10% (5% Per Presentation)

- Considers whether students
  - Speak in a confident, clear and understandable manner that is paced and projected well
  - Have eye contact with their audience instead of with the screen, computer, phone or script

Developing (1)	**Emerging (2)**	**Competent (3-4)**	**Exceeding Expectations (5)**
Little to no attempts at the above	Occasional shows the above	Mostly shows the above	Consistently shows the above

Things To Note About Individual Presentation (10%)

- Given that both presentations will be assessed for Individual Presentation, it is **mandatory** for all students to present in **both** presentations, unless you have a valid reason for missing presentation studio.
- Exceeding the time limit (12 minutes for both presentations) will cause your Group Presentation grade to be penalised

Individual Contribution & Engagement – 15%

- Takes into account students' quality of:
  - Contribution to and engagement in group work
  - Engagement during group consultations and discussions
  - Participation in Question & Answer segments after presentation
  - Participation in class discussions

Developing (1-3)	**Emerging (4-8)**	**Competent (9-12)**	**Exceeding Expectations (13-15)**
Little to no attempts at the above	Some attempts at the above	Active in the above	Shows great depth of thinking and critical engagement in the above

Things To Note About Individual Contribution & Engagement (15%)

- You will be expected to place an end slide to your presentation slide deck showing a table of areas of contribution and which group member(s) contributed, to assist your SL/TA in the grading of Individual Contribution & Engagement:

Area of contribution	Group member(s) involved

Things To Note About Individual Contribution & Engagement (15%)

- **Quality** of contribution in class, in group discussions or in Q&A segments is **more important than** the **quantity** of those contributions
- E.g. one may not speak in every Q&A segment or in every group consultation and class discussion, but when they do, they show evidence of depth of thinking

Peer Review

- Peer Review
  - Mid- Sem and End-of-Sem Peer Reviews are **compulsory** and contributes to **10%** (5% each peer review) of their teammates' grade
    - Liveable Cities Teaching Team will send out a link to students for the Mid-Sem Peer Review: will open on Fri 29 Feb, **due on Fri 6 Mar 11:59pm**
    - Liveable Cities Teaching Team will send out a link to students for the End-Sem Peer Review in Week 13: will open on Thu 2 Apr, **due on Fri 17 Apr 11:59pm**

Criteria to Guide Students' Peer Review Grading

Criteria for Student Behaviour contributing to Good Group Dynamics:

- Communication of intention, deadlines, issues etc in a timely manner
- Effort towards Studio Project
- Time Given for Group Work, Site Visits, Content Creation, Ideation and Discussion
- Communication Dynamics (whether open discussion etc)
- Leadership Qualities, Helping Nature etc

Operating Procedure :

- Students will be asked to give ratings to each other (from 1 to 5 based on the adjacent rubric)
- Average of all the ratings will be considered and will contribute to 10% of the Grade

Peer review scores will be calculated as the average of all scores given by a student's teammates on a software called TEAMMATES

Score	Peer Review Score Criteria
5	• Student went above and beyond to help groupmates in their work • Student ensure that group dynamics was healthy, and there was an open and honest platform for communication and discussion of ideas • Student completed the assigned task in time, and communicated any issues in a timely manner • Student gave excellent amount of time to the group work, joined for many site visits and contributed a lot to content/ideation/discussion
4	• Student completed the assigned task in time, and communicated any issues in a timely manner • Student gave a good amount of time to the group work, joined for many site visits and contributed a lot to content/ideation/discussion
3	• Students sometimes completed the assigned task in time, and was sometimes late in communicating any issues • Student gave adequate amount of time to the group work, joined for some site visits and contributed an adequate amount to content/ideation/discussion
2	• Student rarely completed the assigned task in time, and rarely communicated about the issues being faced • Student gave very little time to the group work, joined few site visits and barely contributed to any content/ideation/discussion
1	• Student never completed the assigned task in time, and did not communicate issues in a timely manner, resulting in uneven workload distribution • Student dedicated negligible amount of time to the group work, did not join meetings/site visits or contribute any content/ideation/discussion

Introducing the Studio Project

Learning Outcomes

By the end of this course, you would be able to understand:

- Appreciate the systems approach taken by Singapore to achieve and sustain liveability outcomes as described in the Liveability Framework developed by the Centre for Liveable Cities

- Understand the considerations taken (for solving complex challenges faced in the development of a liveable city) to develop initiatives, programmes and interventions needed to address problems and/or make improvements in Singapore.

- Apply systems thinking fundamentals to identify & analyse city problems/challenges, propose, evaluate & recommend solution(s)

- Develop teamwork and communication skills through cross-discipline collaboration

Progression Of Studios Up To Presentation 1

Week/Dates	Studio	Activities/Deliverables	Link to previous seminars
Week 3: 27 Jan 2026 (Tue) – 30 Jan (Fri)	Studio 1 (Introduction)	1. Conduct a desktop site research and observation 2. Craft first draft of problem statement based on desktop study 3. Plan and design site visit, survey and interviews	Seminar 2 went through Site Context and Observation
Week 5: 10 Feb 2026 (Tue) – 13 Feb 2026 (Fri)	Studio 2 (Consultation)	1. Downloading data & findings from site visit, survey and interviews 2. Analyse the data & findings Develop first draft CLD, taking into consideration findings from desktop & site studies 3. Update the problem statement	Seminar 3 will go through the Causal Loop Diagram (CLD)
Week 6: 17 Feb 2026 (Tue) – 29 Feb 2026 (Fri)	Studio 3 (Consultation)	1. Refining CLD and problem statement 2. Prepare Presentation 1	-
Recess Week			
Week 7: 3 Mar 2026 (Tue) – 6 Mar 2026 (Fri)	Studio 4 (Presentation)		-

Deliverables for Presentation 1

- Demonstrate understanding of the **demographics and context of the site**
- **Identify problem(s)** for the given site, to be summarized through a **Problem Statement**
- **Analyze the site and the problem(s) to show root causes and relationships between them**
- **Describe relevant stakeholders and communities** connected to the identified problem
- **Give relevant evidence** to support problem identified

- Each team will have to cover these in a **maximum of 12 minutes.**

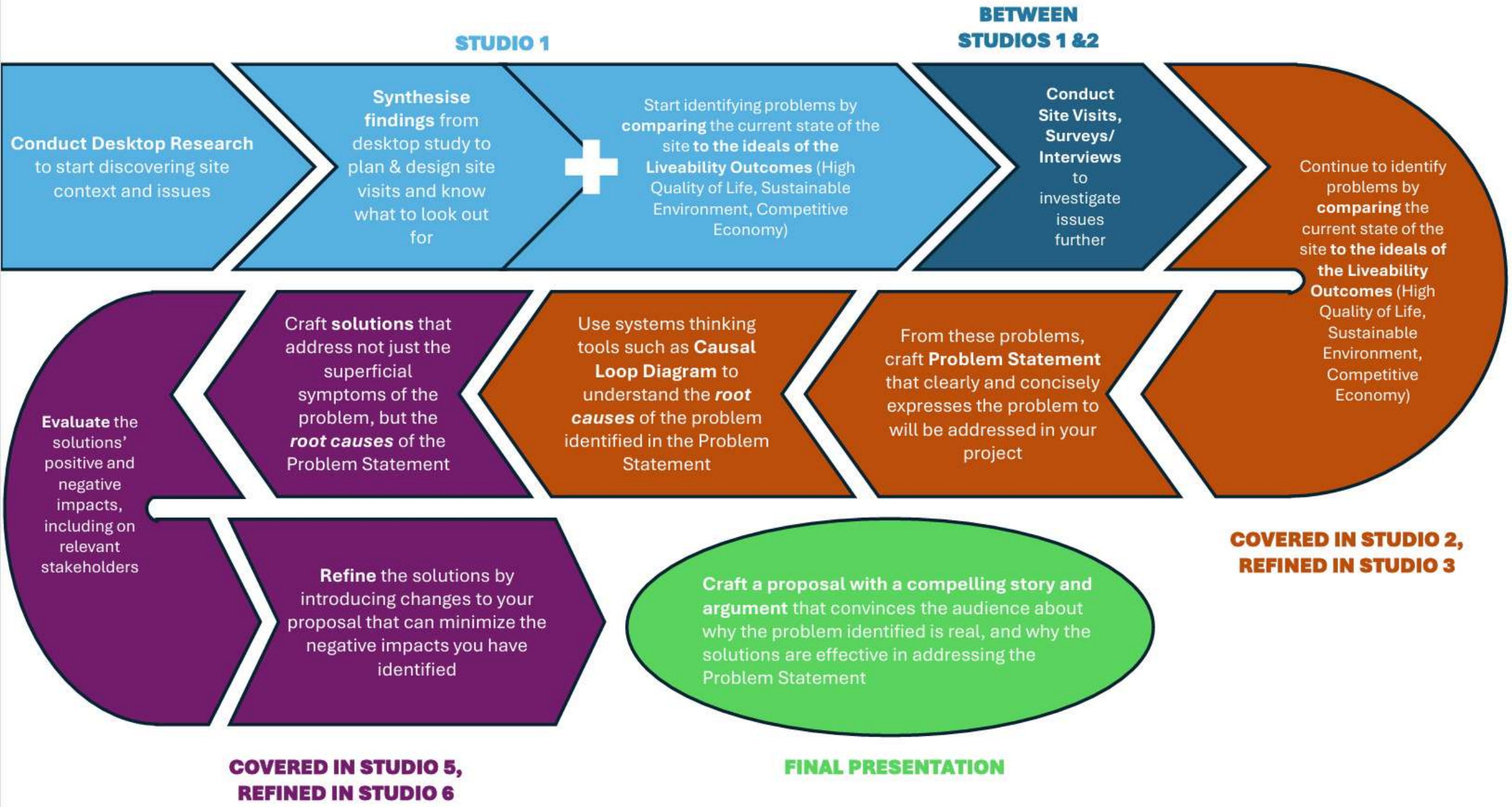
Progression of Studios up to Final Presentation

Week/Dates	Studio/Seminar	Activities/Deliverables	Link to previous seminars
Week 10: 24 Mar 2026 (Tue) – 27 Mar 2026 (Fri)	Studio 5 (Consultation)	1. Refining problem statement, taking into consideration feedback/comment during mid presentation 2. Develop first draft of solution 3. Develop evaluation strategies for the proposed solution	Seminars 5 and 6 will go through considerationg when crafting solutions, and evaluation methods
Week 12: 7 Apr 2026 (Tue) – 10 Apr 2026 (Fri)	Studio 6 (Consultation)	1. Refining solution 2. Refine evaluation strategies for the proposed solution 3. Prepare presentation 2	-
Week 13: 14 Apr 2026 (Tue) – 17 Apr 2026 (Fri)	Studio 7 (Presentation)		-

Deliverables for Final Presentation

- **Brief demographics and context of the site** (specific to the problem)
- **Identification of the problem(s)** for the given site, and summarising it through the **Problem Statement**
- **Analysis of the site and problem(s)** through tools like Causal Loop Diagram
- **Describe the relevant stakeholders/communities** connected to the identified problem.
- **Solution(s)** addressing the root causes of the problem
- **Evaluation of the possible impacts and trade-offs** of the solution(s):
  - **on the relevant stakeholders/communities** (Stakeholder Impact Evaluation) and
  - **on the social, environmental and economic impacts of the solution(s)** [e.g. through S.E.E. Evaluation, Leopold Matrix, Cost-Benefit Analysis or any other evaluation method that shows the social, environmental and economic impacts of the solution(s)]
- **Conclusion**

Each team will have to cover these in a **maximum of 12 minutes.**


STUDIO 1
BETWEEN STUDIOS 1 &2
Conduct Desktop Research to start discovering site context and issues
Synthesise findings from desktop study to plan & design site visits and know what to look out for
Start identifying problems by comparing the current state of the site to the ideals of the Liveability Outcomes (High Quality of Life, Sustainable Environment, Competitive Economy)
Conduct Site Visits, Surveys/ Interviews to investigate issues further
Continue to identify problems by comparing the current state of the site to the ideals of the Liveability Outcomes (High Quality of Life, Sustainable Environment, Competitive Economy)
From these problems, craft Problem Statement that clearly and concisely expresses the problem to will be addressed in your project
Use systems thinking tools such as Causal Loop Diagram to understand the root causes of the problem identified in the Problem Statement
Craft solutions that address not just the superficial symptoms of the problem, but the root causes of the Problem Statement
Evaluate the solutions' positive and negative impacts, including on relevant stakeholders
Refine the solutions by introducing changes to your proposal that can minimize the negative impacts you have identified
Craft a proposal with a compelling story and argument that convinces the audience about why the problem identified is real, and why the solutions are effective in addressing the Problem Statement
COVERED IN STUDIO 2, REFINED IN STUDIO 3
COVERED IN STUDIO 5, REFINED IN STUDIO 6
FINAL PRESENTATION

ADMIN

Introducing Ourselves

Introduction/ Getting To Know You

Please introduce your:

- Name
- Faculty / Major
- One thing that you hope to get out of this course
- One thing that you are uncertain or worried about within the course

Introducing the Studio Project

Sites

- Each team will ballot for one of the following sites:
  - Punggol
  - Tampines
  - Bras Basah-Bugis
  - Chinatown
  - Queenstown
  - Woodlands
  - Jurong

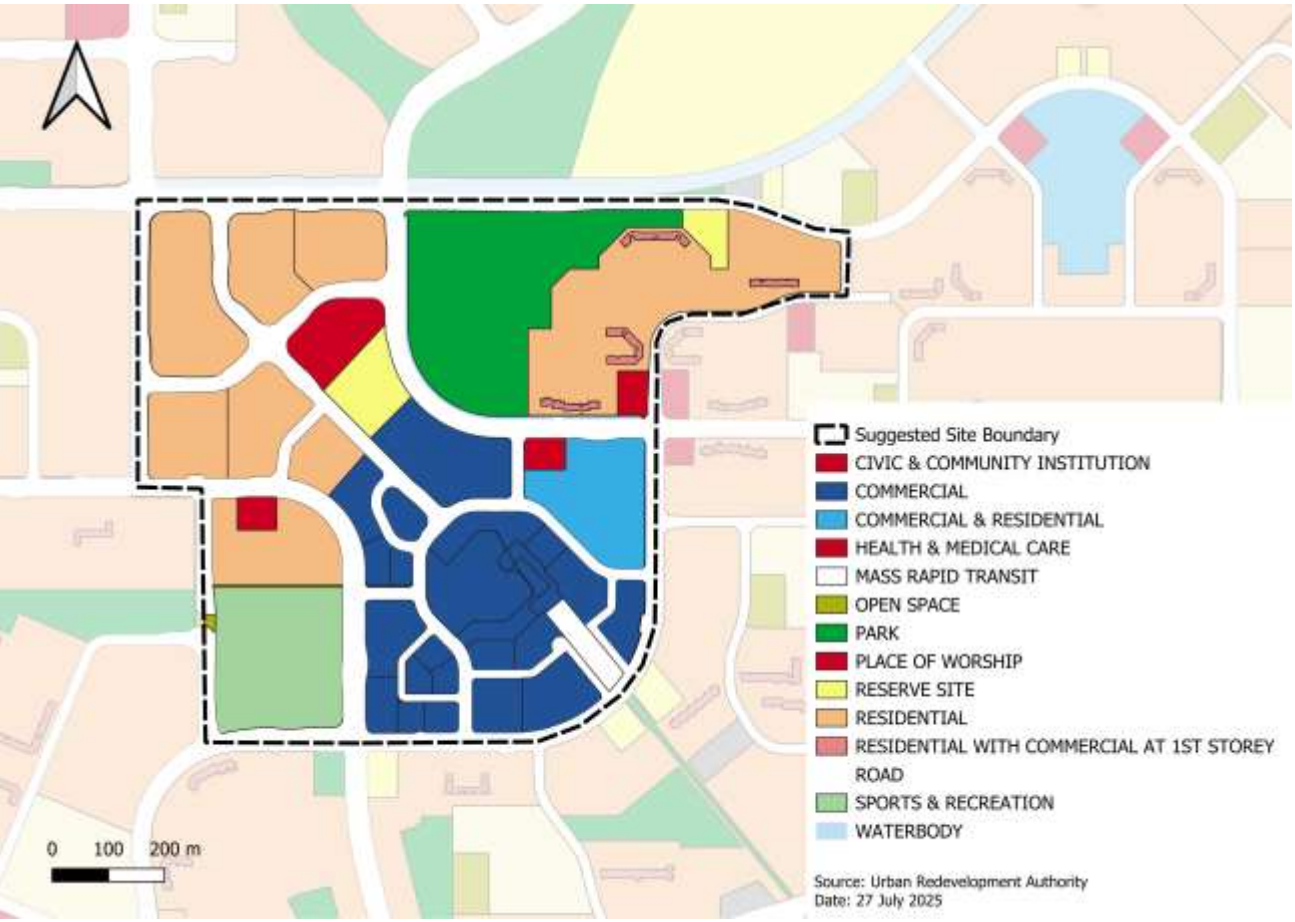
- The site will be the basis for your analysis, problem identification and the context of your proposed solution(s)

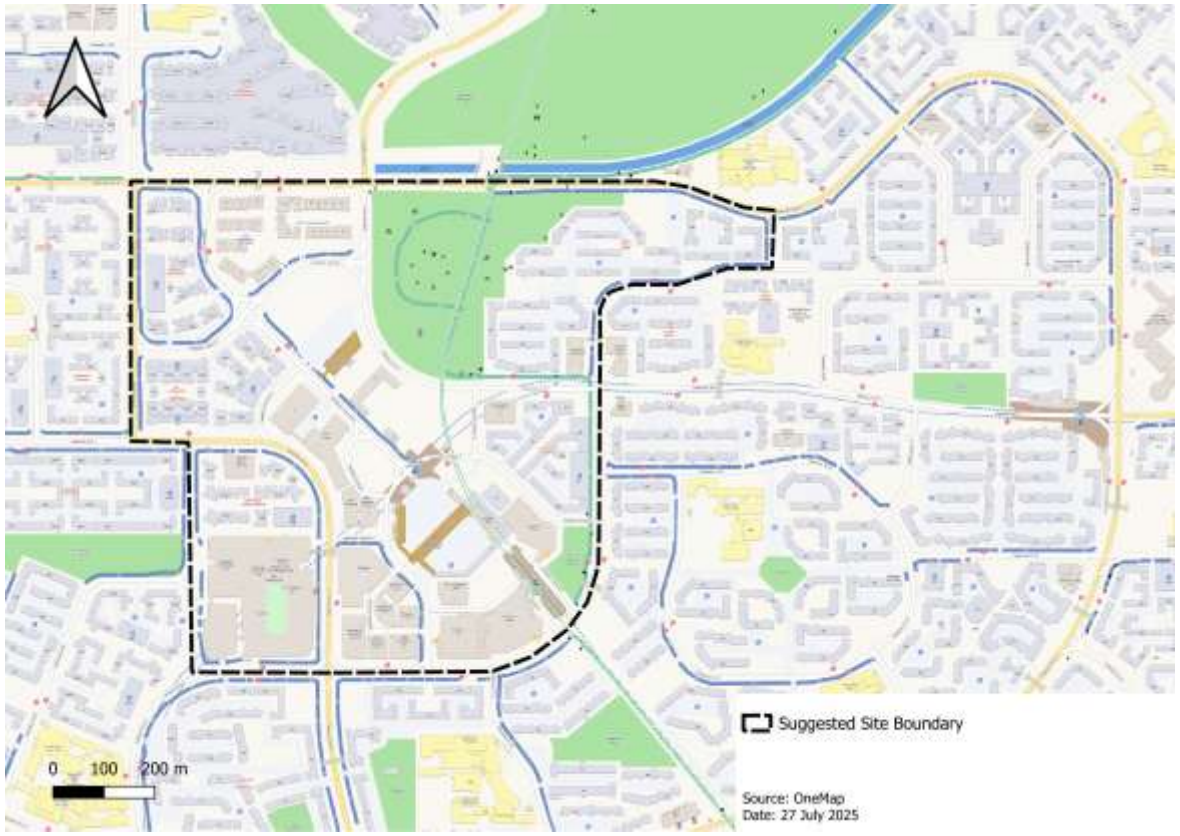
Punggol

Suggested Site Boundary
BUSINESS PARK
CIVIC & COMMUNITY INSTITUTION
COMMERCIAL
EDUCATIONAL INSTITUTION
HEALTH & MEDICAL CARE
OPEN SPACE
PARK
PLACE OF WORSHIP
RESERVE SITE
RESIDENTIAL
ROAD
WATERBODY
0
250
500 m
Source: Urban Redevelopment Authority
Date: 27 July 2025

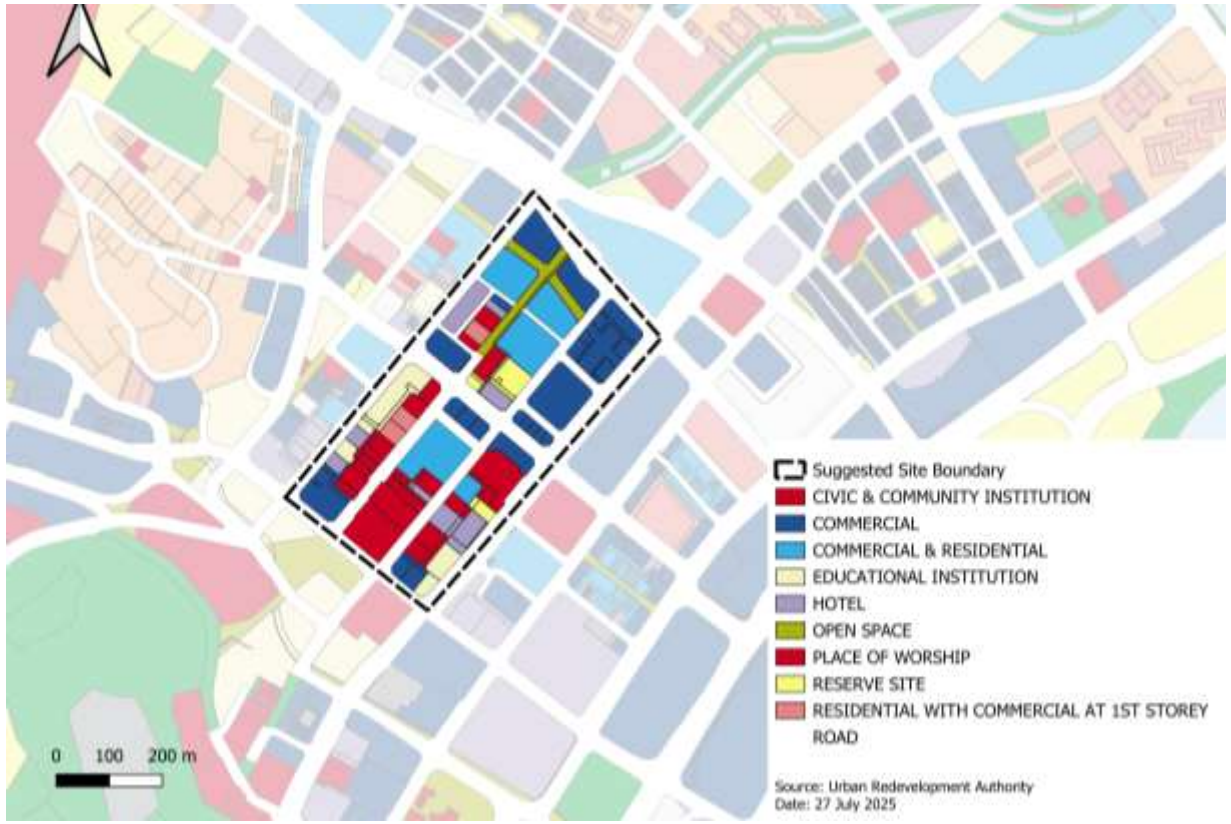
Suggested Site Boundary
0
250
500 m
Source: OneMap
Date: 27 July 2025

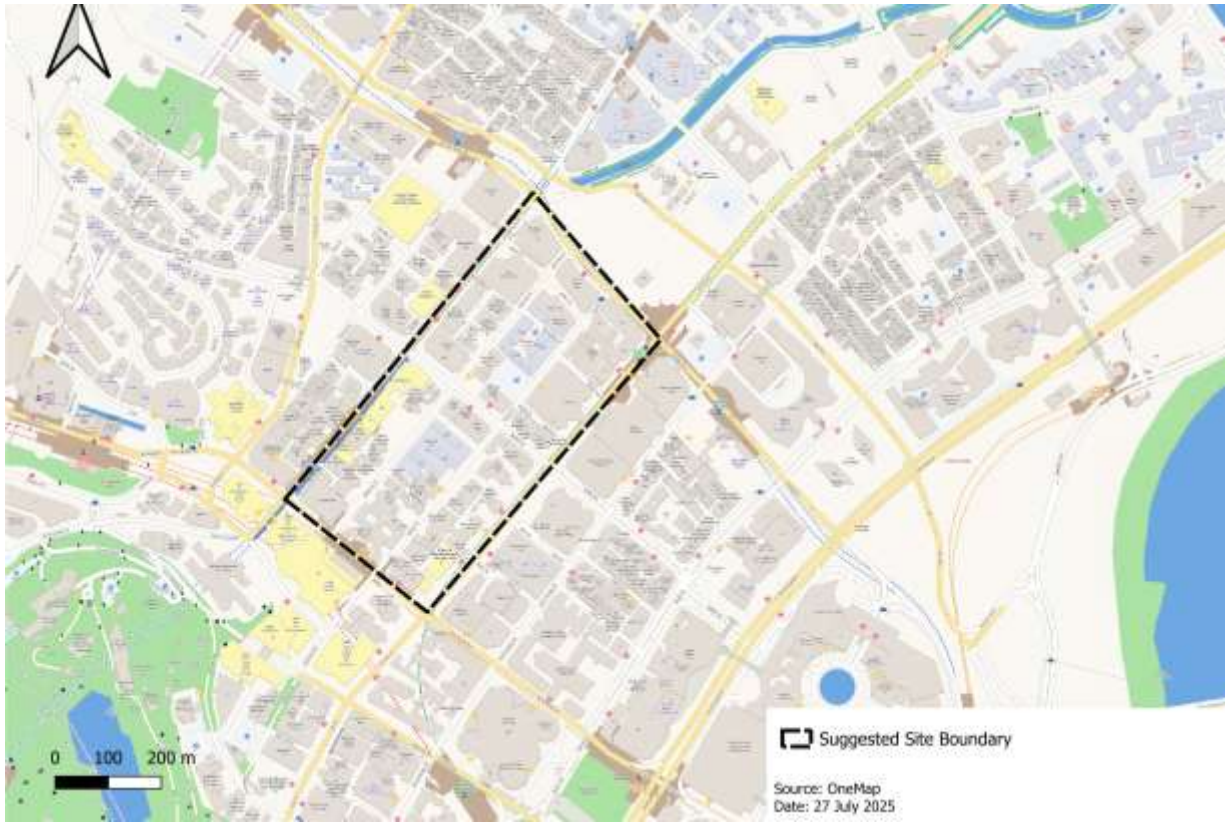
Tampines


Suggested Site Boundary
CIVIC & COMMUNITY INSTITUTION
COMMERCIAL
COMMERCIAL & RESIDENTIAL
HEALTH & MEDICAL CARE
MASS RAPID TRANSIT
OPEN SPACE
PARK
PLACE OF WORSHIP
RESERVE SITE
RESIDENTIAL
RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY
ROAD
SPORTS & RECREATION
WATERBODY
0 100 200 m
Source: Urban Redevelopment Authority
Date: 27 July 2025

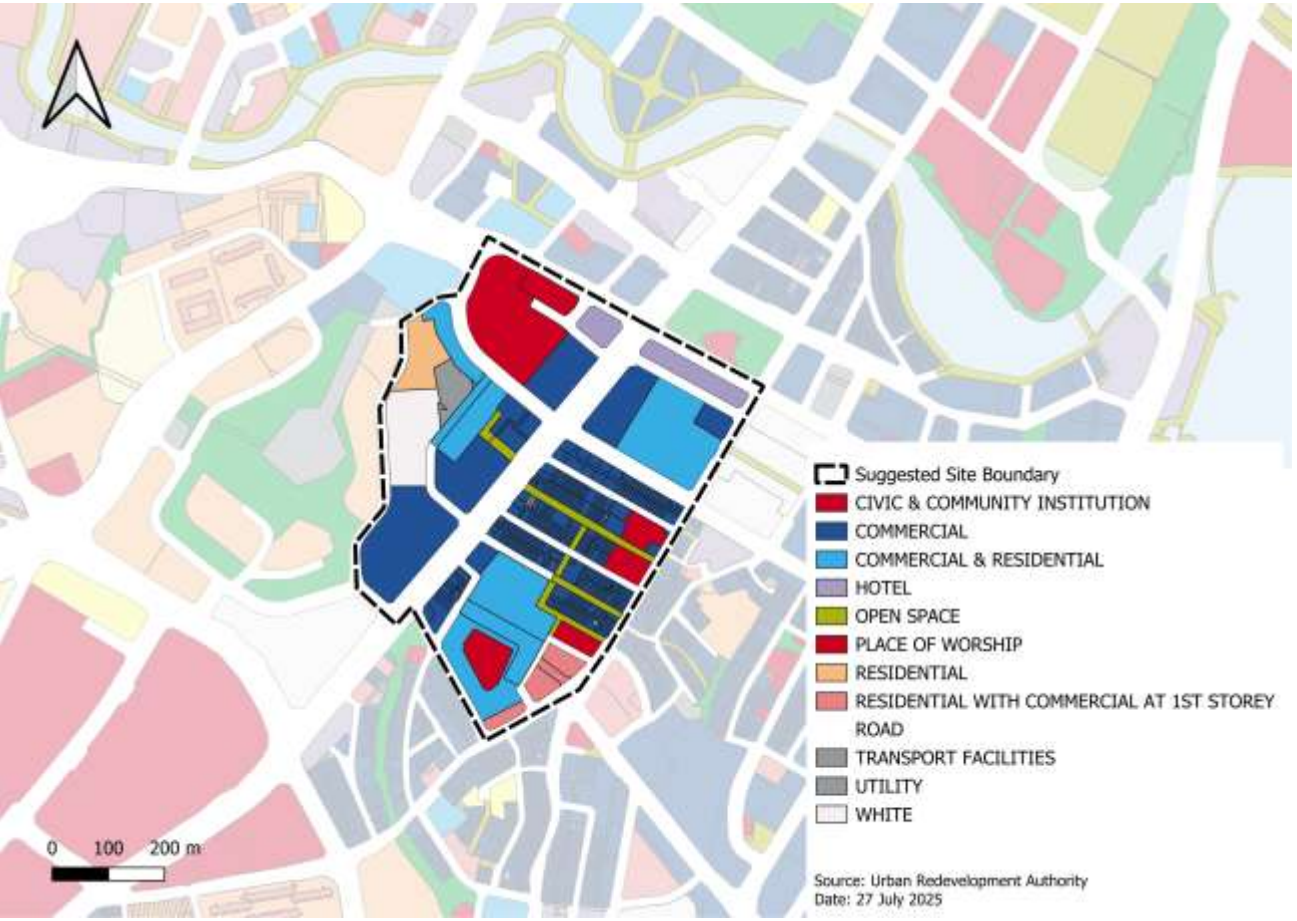

Suggested Site Boundary
0 100 200 m
Source: OneMap
Date: 27 July 2025

Bras Basah-Bugis



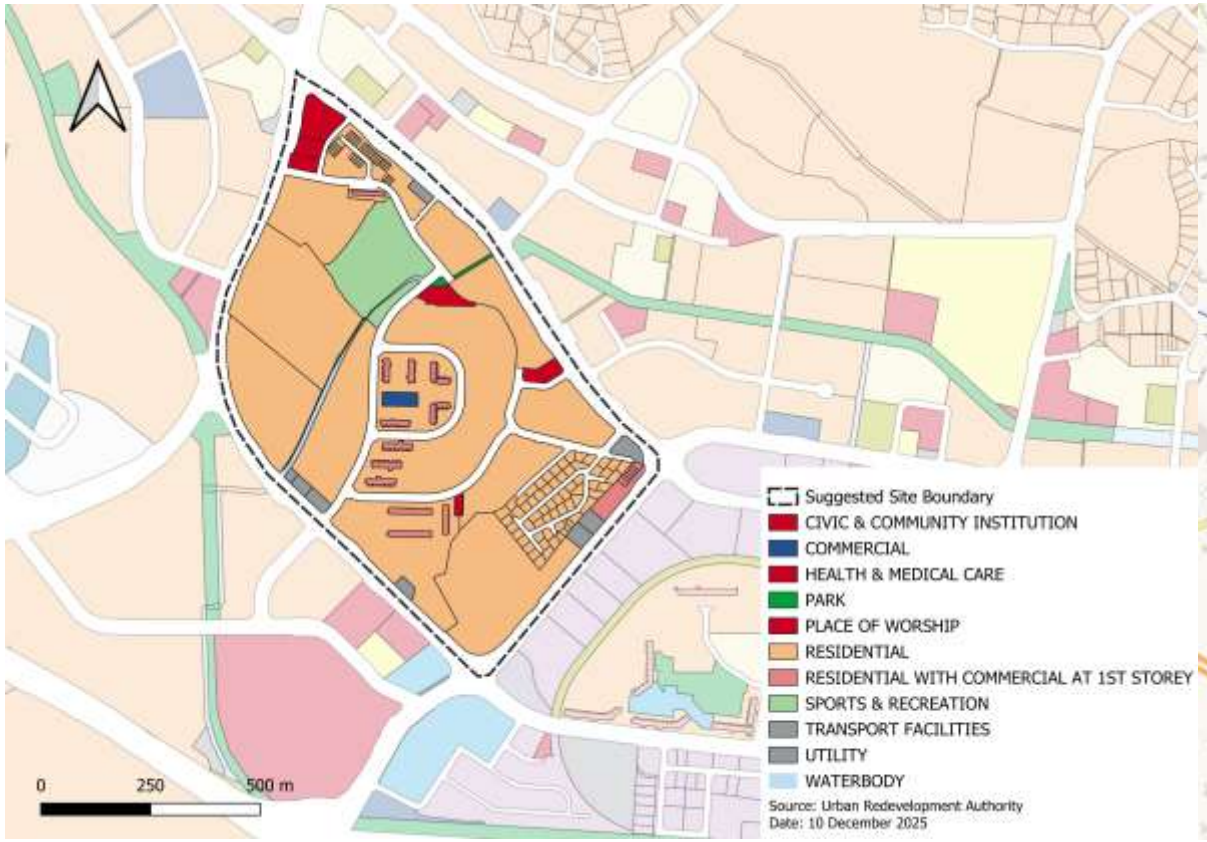


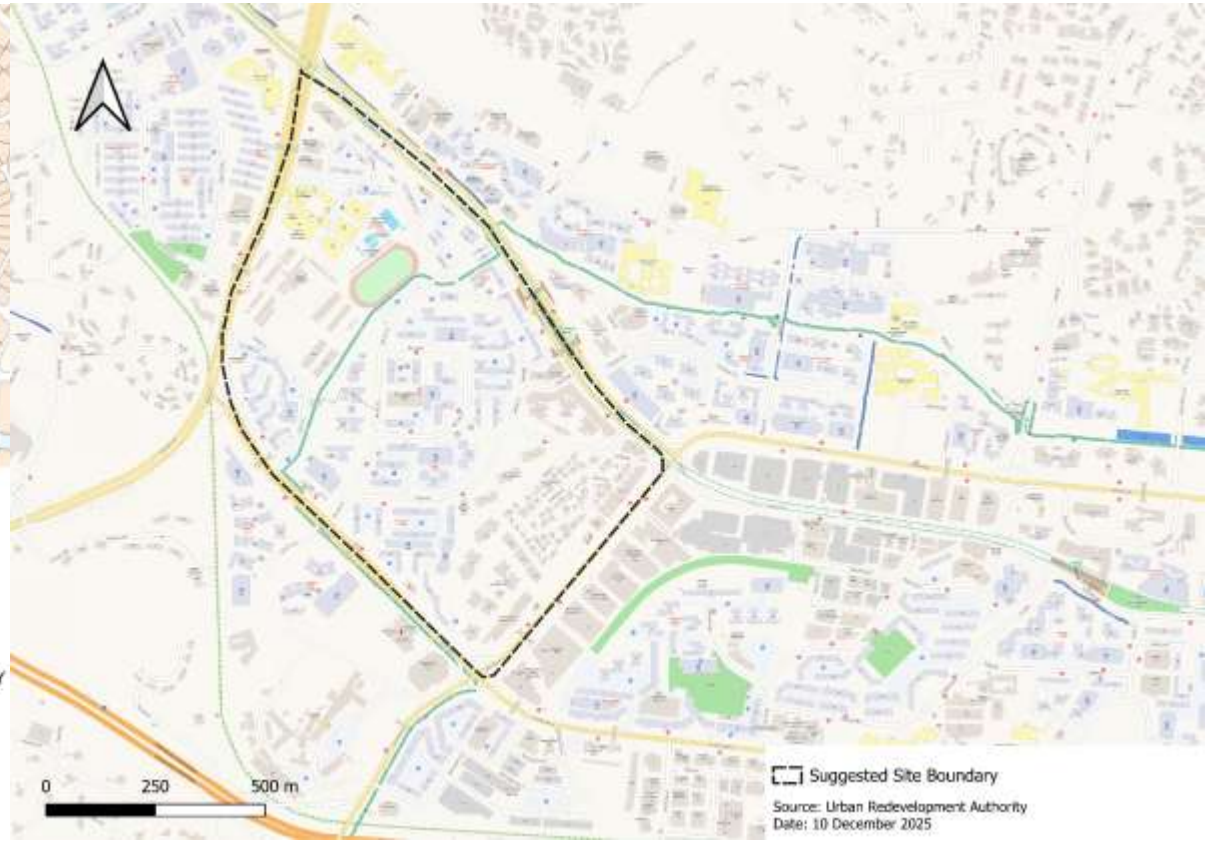
Chinatown


Suggested Site Boundary
CIVIC & COMMUNITY INSTITUTION
COMMERCIAL
COMMERCIAL & RESIDENTIAL
HOTEL
OPEN SPACE
PLACE OF WORSHIP
RESIDENTIAL
RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY
ROAD
TRANSPORT FACILITIES
UTILITY
WHITE
0 100 200 m
Source: Urban Redevelopment Authority
Date: 27 July 2025

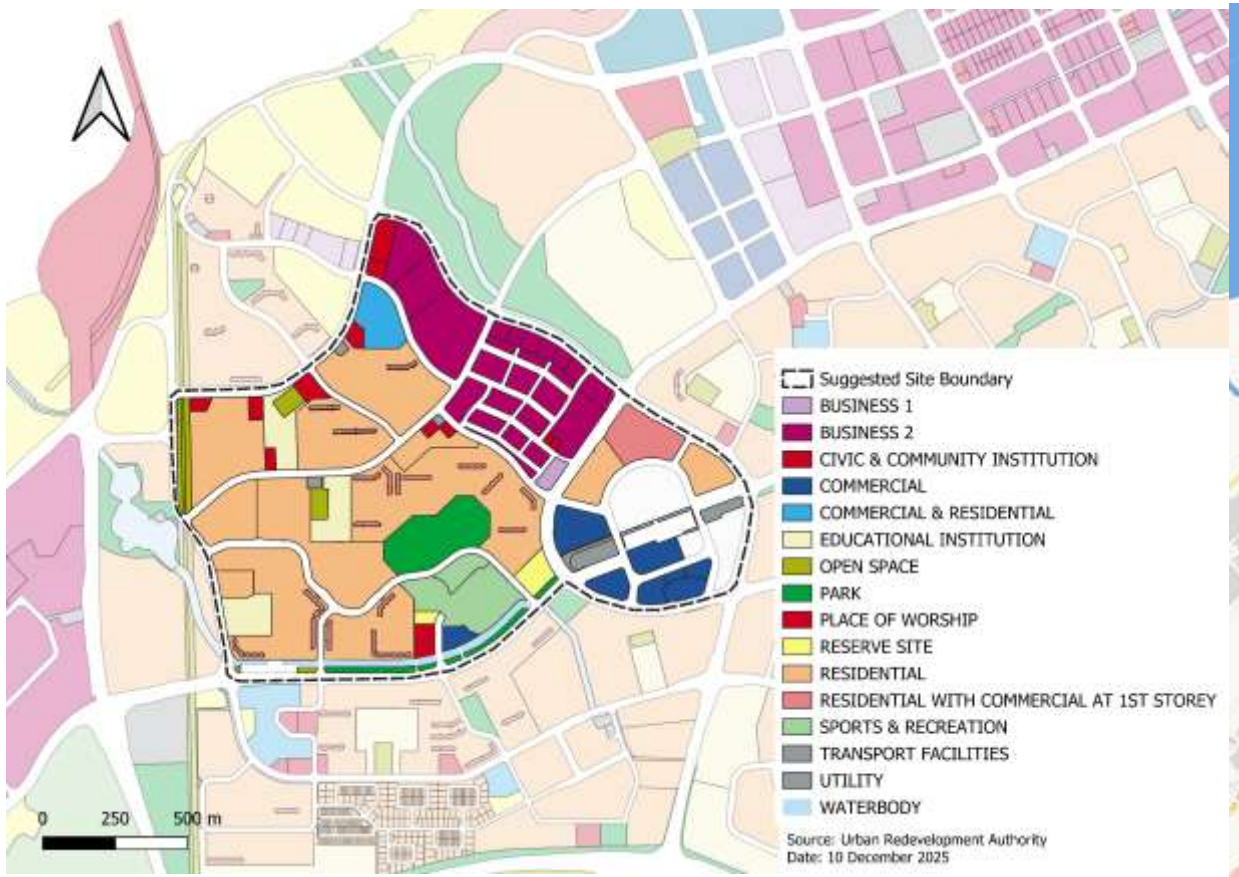
Suggested Site Boundary
0 100 200 m

Queenstown


Suggested Site Boundary
CIVIC & COMMUNITY INSTITUTION
COMMERCIAL
HEALTH & MEDICAL CARE
PARK
PLACE OF WORSHIP
RESIDENTIAL
RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY
SPORTS & RECREATION
TRANSPORT FACILITIES
UTILITY
WATERBODY
Source: Urban Redevelopment Authority
Date: 10 December 2025
0
250
500 m


Suggested Site Boundary
Source: Urban Redevelopment Authority
Date: 10 December 2025
0
250
500 m

Woodlands


Suggested Site Boundary
BUSINESS 1
BUSINESS 2
CIVIC & COMMUNITY INSTITUTION
COMMERCIAL
COMMERCIAL & RESIDENTIAL
EDUCATIONAL INSTITUTION
OPEN SPACE
PARK
PLACE OF WORSHIP
RESERVE SITE
RESIDENTIAL
RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY
SPORTS & RECREATION
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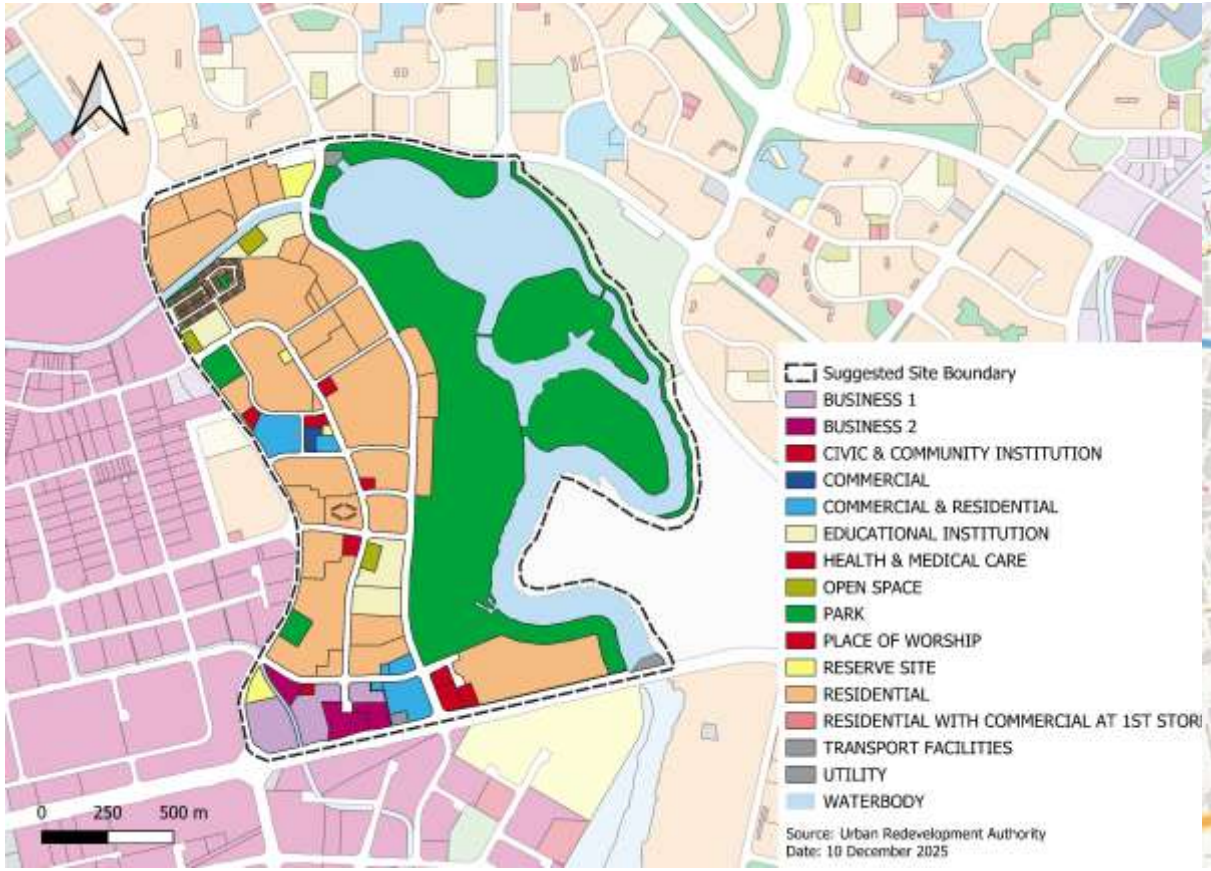
Jurong


Suggested Site Boundary
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EDUCATIONAL INSTITUTION
HEALTH & MEDICAL CARE
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RESERVE SITE
RESIDENTIAL
RESIDENTIAL WITH COMMERCIAL AT 1ST STOR
TRANSPORT FACILITIES
UTILITY
WATERBODY
Source: Urban Redevelopment Authority
Date: 10 December 2025
0 250 500 m

0 250 500 m
Suggested Site Boundary
Source: Urban Redevelopment Authority
Date: 10 December 2025

Now For The Moment Of Truth…

The Wheel of Names (Wheel of Names | Random name picker) will be used.

1. Each team to appoint representative
2. Representative to spin the wheel in the order of the lot drawn!!!
3. Each team may swap or select the remaining sites not selected by other teams.

[2510] 2025/2026 Semester 1

- Home
- Announcements
- Assignments
- People
- Quizzes
- Zoom
- Videos/Panopto
- Grades

All Site Boundaries.pdf

Site Boundary Images

URA SPACE (Detailed Land Use Plan)

OneMap (Detailed Street Map)

Additional Resources

Additional Studio Resources.pdf

Research Resources for your Projects (HELPFUL FOR UNDERSTANDING SPECIFIC ISSUES)

Videos (Skillsets)

Boundaries of all five sites are also available on CANVAS!

[2510] 2025/2026 Semester 1

- Home
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All Site Boundaries.pdf

Site Boundary Images

URA SPACE (Detailed Land Use Plan)

OneMap (Detailed Street Map)

Additional Resources

Additional Studio Resources.pdf

Research Resources for your Projects (HELPFUL FOR UNDERSTANDING SPECIFIC ISSUES)

Videos (Skillsets)

These are additional resources on:

- How to conduct site visits
- How to conduct research
- Starter resources on specific issues

Managing Team Dynamics

How Can We Work Well In Teams?

- The Liveable Cities Studio Project is a team project and places you in teams consisting of fellow students of different majors
- Collaboration in teams provides opportunities (enriching one another with your unique perspectives) and challenges (conflict due to disagreements, different expectations and working styles)
- **How can you help your team to have a productive and even enjoyable time working together?**

The Importance Of Asking And Listening

- In teamwork, there is the potential for misunderstandings and misinterpretations of each other's views
- Additionally, when one or a few members dominate the conversations, others can feel that their views do not matter, which could even cause them to decide that putting in effort isn't really worth it
- **Hence, it is important to engage in asking questions and in reflective listening**, to properly understand and clarify our understanding of each other's views

Tips For Better Asking And Listening

- The practice of **reflective listening** can help you to understand and to let others feel understood
- There are a few ways to go about it:
  1. Summarise and paraphrase
     - "So what I am hearing is that…[paraphrase what you heard from your peer in your own words]. Is this correct?"
     - This gives the other person the chance to check your understanding, how accurate is it
  2. Clarify
     - "Tell me more"
     - In saying this, you acknowledge that you may not have fully understood what the other person said, and give them a chance to elaborate further to enhance your understanding

Dialogue vs. Discussion

- What we just went through, about the importance of asking questions and listening, is important for engaging in **dialogue**
- This is different from **discussion**, where the aim is to come to a decision on something
- Both are necessary
- Dialogue more useful when things are uncertain at the start – sometimes it may be more important to slow down and seek to understand one another's views first, and not rush into making judgments about what's the best way to move forward
- Discussion more useful when a decision needs to be made to move forward, and this can be important too
- Hence, there is no hard and fast rule as to when to dialogue, or when to discuss – as long as it is purposeful

Creating a Collaborative Plan

- A deck of slides should have been sent to your email
- Use it as a guide to what you can discuss with your team about and come to a common understanding of:
  - What are your individual goals? Is there a common team goal that everyone can agree upon?
  - Be frank with what you are looking to get out of this course e.g. do well, SU this course, etc. – this will be helpful in setting realistic expectations among your teammates
  - How will your team communicate?
  - What are some boundaries that you would like your team to respect? (But be open to compromising where necessary)

Why Create a Collaborative Plan?

- Needed to:
  - Ensure a common purpose among all team members
  - Smooth coordination in teamwork
  - Build trust
  - Build high team cohesion
  - Perform better as a team

Planning Together On Goals

Purpose	**Team Goal** *What do you as a team hope to accomplish?*	
	Individual Goals *What do you as an individual hope to accomplish?* *What are some potential competing priorities?*	

Clarify What Are Your Goals For This Course

- Grade goals?:
  - Do you just want to pass or SU the course?
  - Do you want to excel in the course and do very well?
- Other goals?:
  - “I want to learn more about cities because it contributes to my future career path.”
  - “I just want to get this course over and done with.”
- Competing goals?:
  - “I am able to devote my time and energy to this course.”
  - “I need to spread out my time and energy among other courses.”
  - “I need to spend time with my family.”
  - “I need to spend time and energy at this part-time job I’m having.”
- Knowing each other’s differing individual goals can be helpful: you know what you can expect of one another and can shape your collaboration style according to that
- Even better: Is there a common goal that your team can work towards?

Planning Together On Communication Methods And Protocols

Coordination	**Communication** *How will you run your meetings? In-person or Online? How often?* *How will you communicate when not together?* *How will you make decisions and document them?* *What do you do when you cannot agree with each other?*	

Managing Communication

How will you run your meetings? In-person or Online? How often?

How will you communicate when <u>not together</u>?

How will you make decisions and document them?

What do you do when you cannot agree?

Managing Meetings

Biggest omission

Failure to establish regular meeting times beforehand

Choose an approach to decision making

Consensus?

Majority rules?

Deference to leader?

Conflict Resolution

Clarify the disagreement

Review goals

Seek compromise

Seek root cause

Assign a mediator or escalate?

Use Collaborative Planning Template

Trust	**Psychological Safety** *What are some boundaries and ground rules?* *When can you expect replies? When would you not reply?* *How early in advance must we notify others when we are unable to fulfil our commitments?*	

Boundaries

- It is important to clarify what boundaries we have
- These boundaries could include:
  - Is it ok to message each other at the middle of the night?
  - Is it ok to ask for things over the weekend?
- We respect each others' boundaries
  - Unless it is critically important and a compromise needs to be made, it may be counterproductive to force others to agree with our way of doing things
  - Create an environment where it is alright for someone to say "no" (with reason) without feeling that they will be judged or condemned
  - But do not abuse this either: saying "no" for no reason is not helpful
  - Ask one another when you are not sure or clear about another person's boundaries
  - Each person also has the responsibility to voice out their boundaries: some boundaries may not be obvious to others unless said and explained

Reliability

- Is about doing what you say you'll do
  - Requires us to stay aware of our competencies and limitations
  - Requires us to also be aware of what we can actually do vs. what you hope you would be able to do
  - Requires us to be aware of our competing priorities, and how much of our time and energy we can devote to a particular piece of work
  - Don't over-commit when you are clear that you will not be able to handle it due to the above pointers
- Reliability is subjective
  - E.g. How much lead time is needed if one cannot deliver on commitments or responsibilities? – this question can have different preferred answers for different people
    - 2 hours or 12 hours or 1 week?
    - Like boundaries, it is important to clarify what each person means by reliability

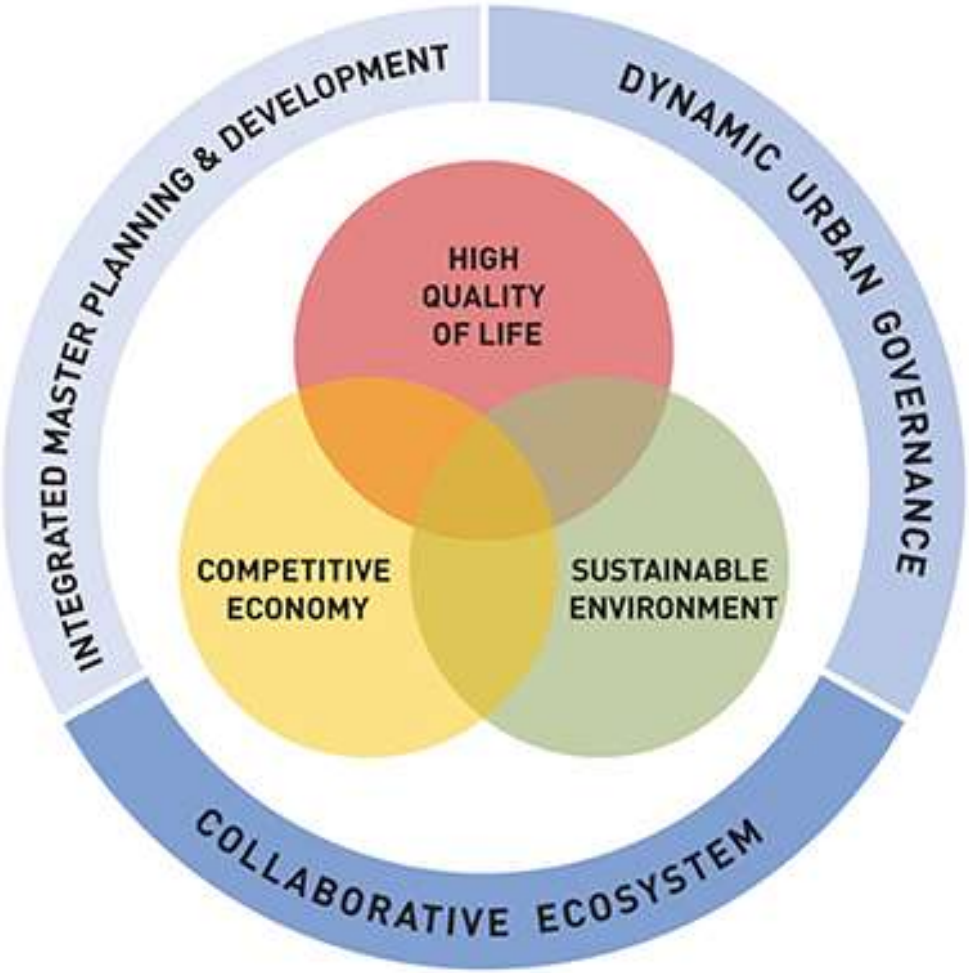
ASSIST

Using LF As A Lens to Identify Problems

Liveability Framework

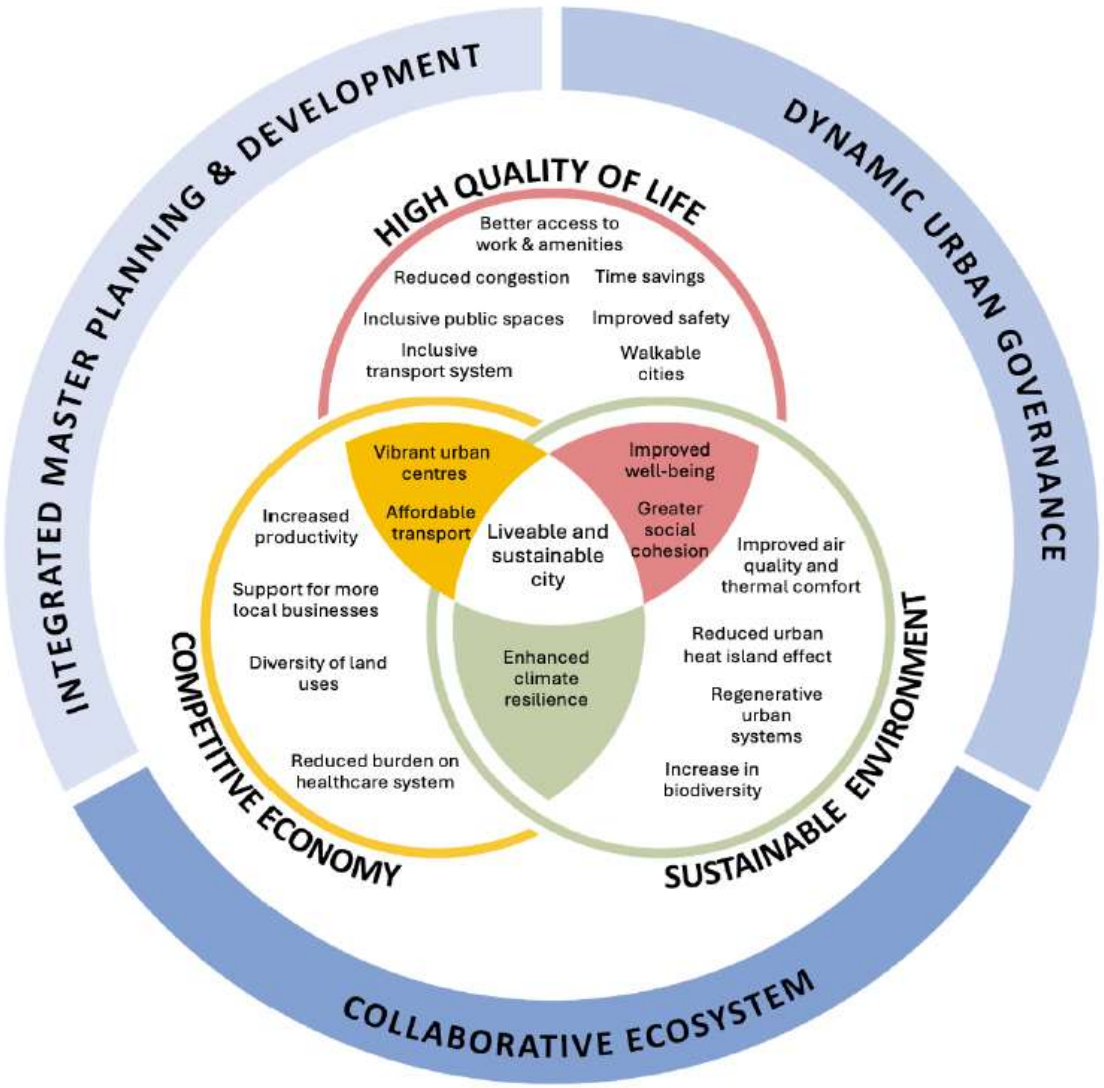
- Recall the Liveability Framework and the three Liveability Outcomes (High Quality of Life, Sustainable Environment and Competitive Economy) that was introduced in Seminar 1

- How can these be used as a lens to view urban problems at your site?



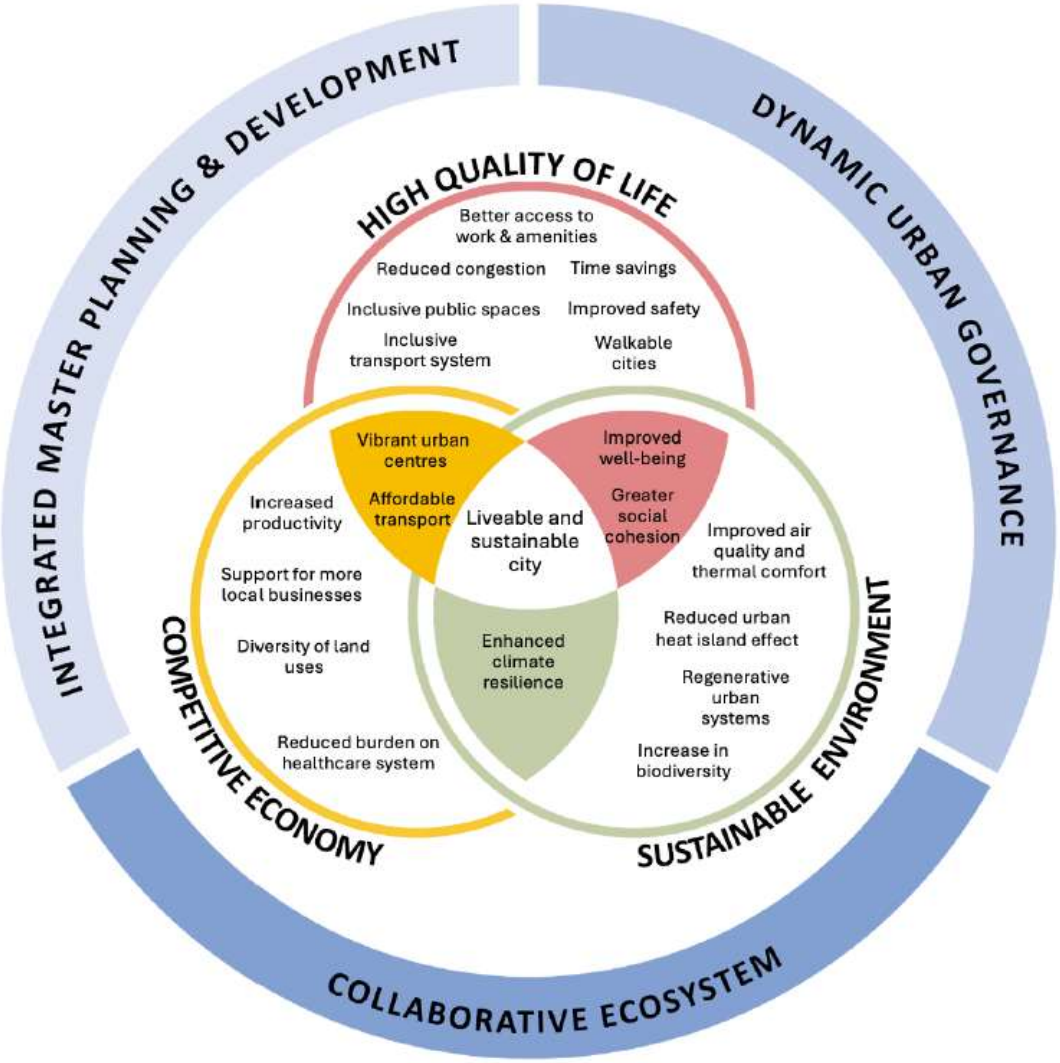
High Quality of Life

- When you look at issues using this lens, they are typically related to
  - Fundamental aspects of liveability that are criticial to keep residents safe and healthy and allow them to live comfortably
    - Eg: right infrastructure, utilities and public services to meet day-to-day needs
  - Higher-order needs of residents
    - Eg: sports, recreation, arts and culture, social and community assistance



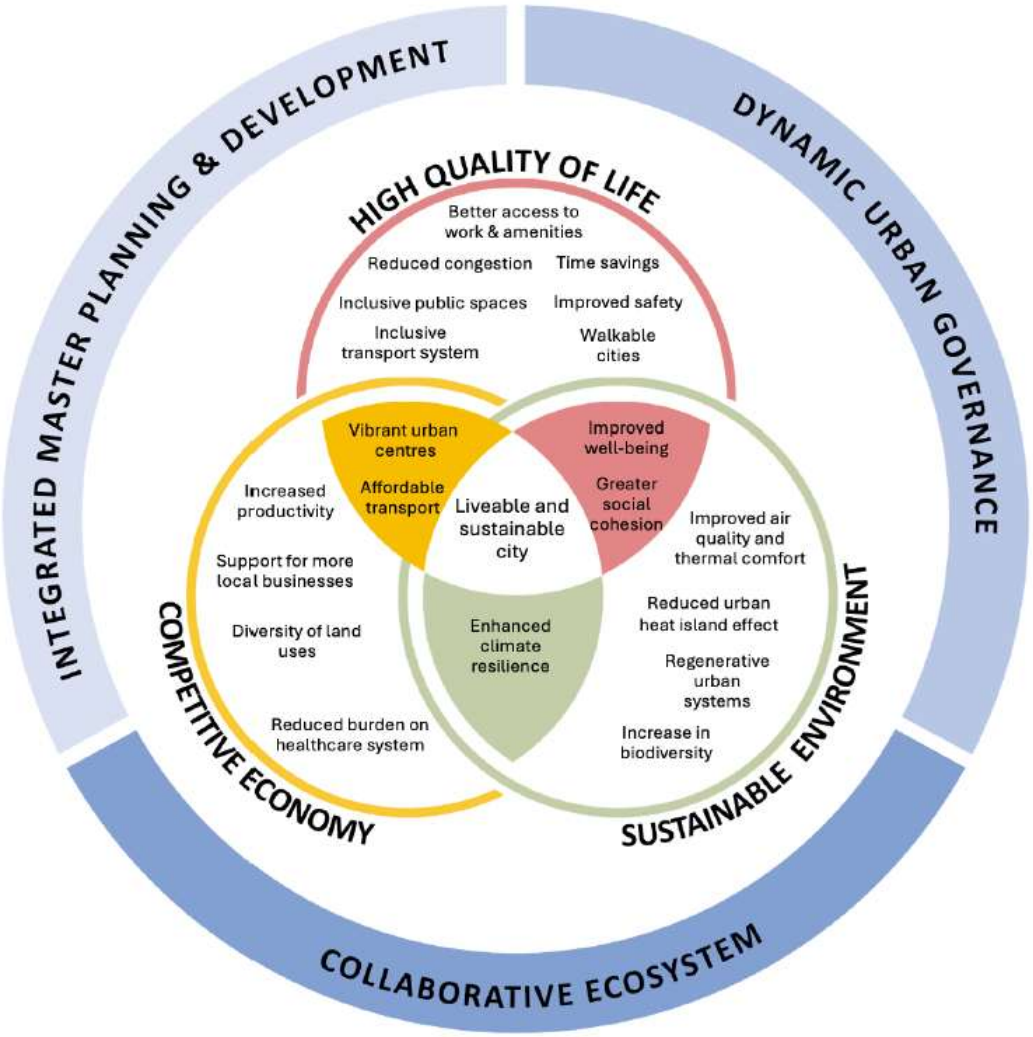
Sustainable Environment

- When you look at issues using this lens, they are typically related to
  - Environmental protection
  - Creating a green, liveable and sustainable environment
  - Responsible usage of natural resources
  - Maintaining resource, food and energy security



Competitive Economy

- When you look at issues using this lens, they are typically related to
  - Ways Singapore can create a safe stable and conducive environment for businesses no matter big or small
    - Eg: allocation of land & infrastructure for industrial use, efficient transportation networks
  - Implemented policies to help businesses expand overseas



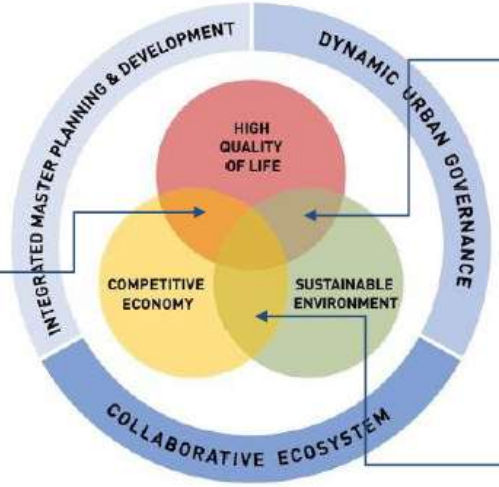
How this Leads to Identifying ~~Problems~~ Urban Issue in the Site

- A problem of your site can be defined as the gap between the current reality of your site and the ideals of the three Liveability Outcomes (just covered)
- To identify potential Problem Statements, ask yourself:
  - How far does my site achieve a High Quality of Life for all?
  - How far does my site have a Sustainable Environment?
  - How far does my site support a Competitive Economy?
- Your answers to these questions could raise potential problems that can be crystallised into a Problem Statement, which we will cover next week

Considerations

Intersections: Tensions & opportunities

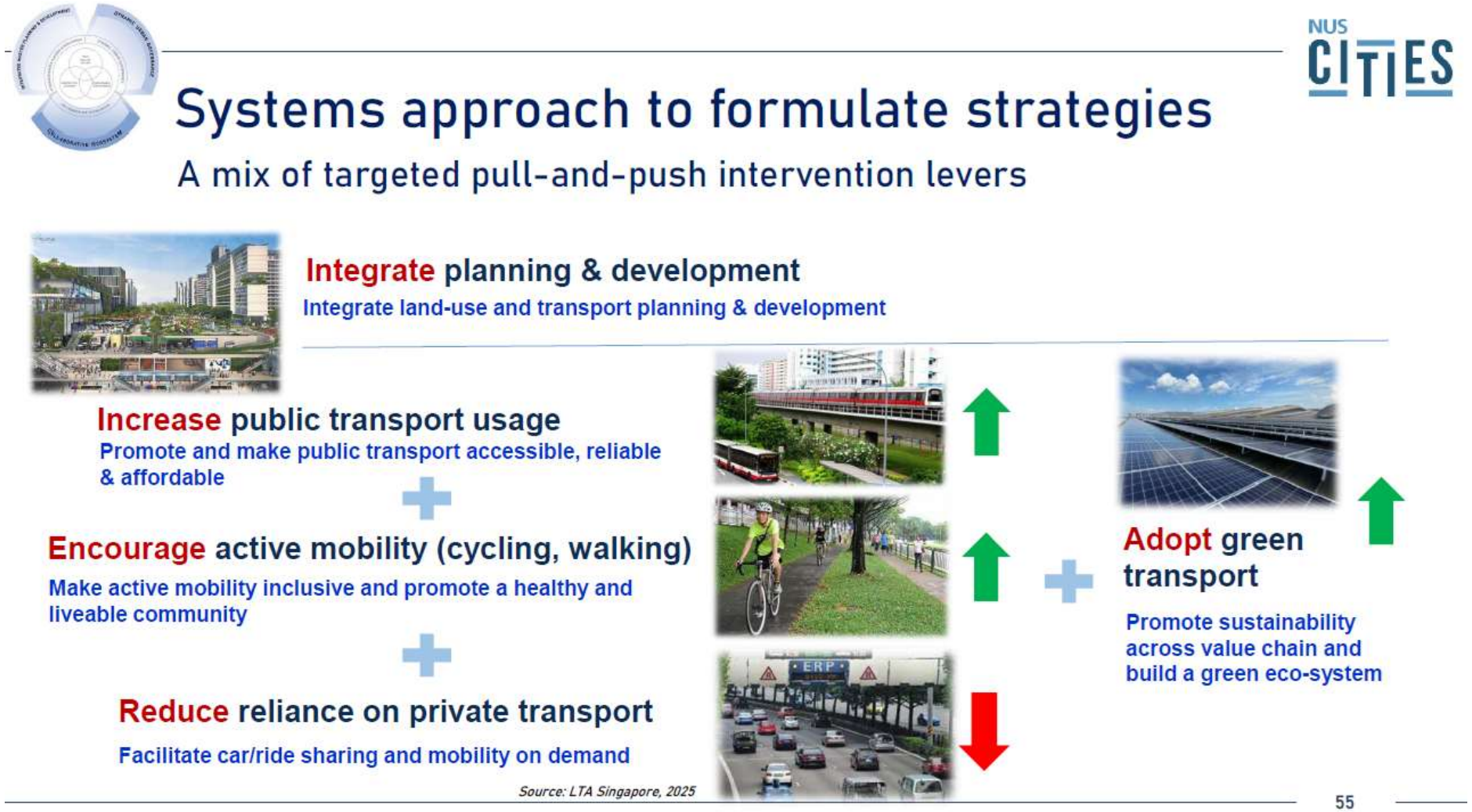
- Redevelopment vs Conservation/Heritage
- Workforce productivity vs Health/Leisure
- Connectivity to support businesses vs Spatial quality



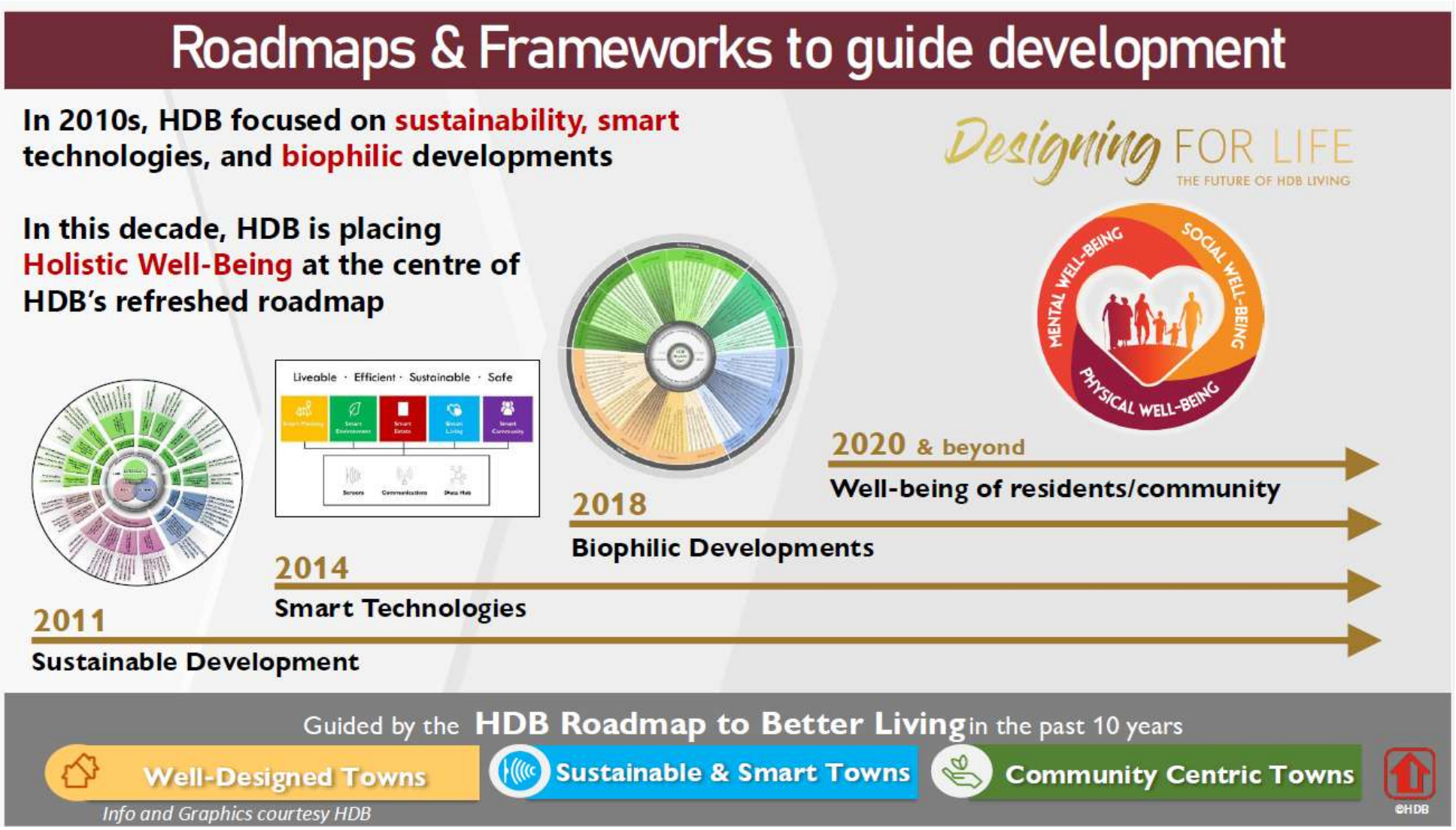
- Greenery vs Development
- Human – Wildlife conflicts

- Costs to businesses
- Cost of living

Considerations



Considerations



Today's Learning Objectives

Today's Objectives

1. Understand the importance of **contexts** in understanding site
2. Understand what to **observe** in a site visit to build your understanding of the site
3. Establish consensus related to **team works** and group dynamics

Site Visits

Doing Your Site Visit

- Welcome to the next step of learning more about your site – doing your site visits!
- To make the most of your site visit, consider the three-step process below:
  - What to do **before** your site visit
  - What to do **during** your site visit
  - What to do **after** your site visit

Before Your Site Visit

1. **Research the district/site**
   - Through desktop research and online platforms (eg Google Earth), **identify characters and areas of interest** to look out for during site visit – e.g. transportation, housing, green spaces, population ageing, historical significance, economic activities

2. **Plan your data collection method and design questions in advanced**
   - Have a brief idea of what data you want to collect
     - For example: if you decide to do an interview, identify who you want to interview and why (e.g. demographic profiles of interviewees), what kind of questions you want to ask

Tip for Effective Data Collection

- Keep a **fieldwork journal** that you will use to take notes during site visits/group discussions/class
- Helps you track changes in your observations and analysis as you (re)visit the site and progress throughout the semester

During Your Site Visit

1. **When you first reach site:**
   - Note time of your visit as **when you visit area may significantly affect your findings**
   - Note **first impressions** and overall atmosphere of place
   - Observe **physical layout** of place, e.g. buildings, streets, parks, infrastructure, human traffic and demographics, availability of sheltered walkway
   - Record which **demographic** is there e.g. who is there, what are the age groups, ethnicity, possible occupations

2. Throughout your time at the site, take plenty of **detailed notes and pictures/videos** of key areas and issues that you may see
   - If you're unsure on what to observe, take photos of it anyway! You can always revisit it later on and notice new things you may not have seen before

3. Record **questions that may arise** for you to address later
   - There are no right or wrong questions!

What Should I Look Out For During A Site Visit?

1. **Observe the hardware / space**, look out for things such as:
   - **Infrastructure** e.g. sidewalks, sheltered walkways, graffiti/street art, dark alleys, poorly lit areas, ramps/handrails
   - **Housing** e.g. types of housing, housing conditions, occupancy rates,
   - **Public spaces** e.g. availability and conditions of parks, recreational areas, cleanliness, usage
   - **Community areas** e.g. community centers, polyclinics, libraries
   - **Commercial areas** e.g. types of businesses, foot traffic, economic activity
   - **Traffic and transportation** e.g. flow of traffic, congestion points, availability/ affordability of transport, peak times for traffic, busy vs quiet areas, unsafe crossings
   - **Signage and Wayfinding** e.g. clarity and effectiveness of signs and directions
   - **Environmental aspects** e.g. green spaces, pollution levels, waste management, presence of litter, noise and air pollution, air quality

What Should I Look Out For During A Site Visit?

Use the 5 Ws Tool to guide you:

- **Who**? Stakeholders
- **What** is happening? Behavior/activities of people in area and problems they face
- **Where**? Provide context – What you see in your living environment impacts how you read/understand the urban context
- **When**? Time matters, when and how long you visit the site will give you different observations; e.g. CBD on a weekday lunch period VS CBD on a weekend lunch period
- **Why**? What are the reasons for the behavior/activities of people causes of problems that they face?

What Should I Look Out For During A Site Visit?

2. Engage in detailed observations of software/human interactions/social relations

- **Observe interactions** among people e.g. their behaviors and activities, what they are doing
- **Demographic** details e.g., age group, cultural diversity
- Pay attention to accessibility issues: are there facilities for older people and persons with disabilities?
- If possible, conduct **interviews/surveys** with the community members
  - Always ask for consent before proceeding with the interview/survey
  - Ask open-ended questions

After Site Visit

1. **Review and reflect**
   - Review notes and reflect (make sense of the collected data) on observations and findings: what **possible problems** did you identify?
   - **What** questions did you have when you were on the site visit? **How** can they contribute to formulation of project?

2. **Analyze**
   - Consider downloading and making sense of your learnings again!
     - Categorize identified problems into **key categories or themes**
     - Identify patterns and connections between different categories and/or themes
     - Identify **linkages between data collected or findings from site visit to problem statement**(s) you identified

3. **Summarize**
   - Summarize **key findings and corroborate with evidence** e.g., pictures, videos, sketches

Understanding The Site Context

Understanding Site Context

- In your studio project, you will have to identify a problem which can be found in your site that you would like to address
- Before going to the stage of identifying a problem, you will need to do some research to understand better the context of your site
- The main types of context you can look into to understand your site better are:
  1. Socio-economic
  2. Historical
  3. Spatial
  4. Governance

Let us look at each of these in turn

Socio-economic Context

- Understanding the socio-economic context of your site is about answering the following questions:
  - What are the different social groups that live, work and play in your site?
  - Which social group is predominant in your site?
  - What are the needs/aspirations of the social groups of your site?
  - In what ways does your site provide for or not provide for these needs/aspirations?
- Social groups can be categorised according to:
  - Age (elderly; middle-aged adults; young adults; teenagers; children)
  - Socio-economic status (High income/high wealth to low income/low wealth)
  - Gender
  - Family structure (e.g. single mothers, young families with working parents, families taking care of elderly parents, singles)
  - Others?

Socio-economic Context: Universal Needs

- Some needs could be considered universal, crossing different social groups:
- These universal needs mean that there are certain characteristics of the city/neighbourhood that are important for all social groups, for example:
  - Health: provision of healthcare (polyclinics, hospitals, pharmacies), spaces and programmes for exercise (gyms, fitness corners, fitness classes), provision of healthy food options (at the supermarkets, restaurants, hawker centres)
  - Safety and Security: proximity of police station, provision of lighting, community watch groups, street design, enforcement of traffic laws
  - Social support: provision of public spaces to meet, recreation areas to engage in activities together with friends and family, social services, eldercare, community initiatives and activities etc.

Maslow's hierarchy of needs

Socio-economic Context: Unique Needs

- However, there are some needs which are unique to a particular social group, which means that the city/neighbourhood needs to be planned/designed for their needs, for example:
  - For elderly persons who are less mobile, they may need ramps and wheelchair friendly infrastructure, access to healthcare through tele-services or mobile doctors, access to caregivers, etc.
  - For families with working parents, they may need childcare centres, children play areas, proximity to grandparents as potential caregivers, etc.
  - For those of low income/low wealth, they may need access to social workers and financial assistance, access to healthy food at lower cost, etc.

Understanding The Site Context (Continue)

Historical Context

- Planning decisions and implementation of infrastructure, programmes, and policies in the past were made for particular reasons in the context of that time period
- Those reasons may no longer be relevant, given changes that have happened since then
- Some of the possible shifts that may cause plans, infrastructure, programmes, and policies to become less relevant include:
  - Shifts in demographics
  - Shifts in city-wide/national priorities
  - Shifts in tastes and preferences

Historical Context: Shifts In Demographics

- E.g. A neighbourhood or town that was planned for a much younger population in the past may not be equipped to serve an ageing population
- Singapore has already begun to adapt its neighbourhoods through programmes and infrastructure such as:
  - HDB's Enhancement for Active Seniors (EASE) Programme, which provides retrofit features such as grab bars, slip-resistant treatment to existing floor tiles, and ramps in the homes of the elderly
  - LTA's Silver Zones institute specific traffic policies, road infrastructure and other design features to encourage motorists to slow down and pedestrians to exercise caution
- While such efforts have been started, you can still ask whether your site still contains the legacies of past decisions that did not take into account future needs

Historical Context: Shifts In City-wide /National Priorities

- Sometimes urban infrastructure, designs, policies and programmes prove to be inadequate because they were planned with less information in the past, or where circumstances have changed e.g. climate change

"Air conditioning was a most important invention for us...The first thing I did upon becoming prime minister was to install air conditioners in buildings where the civil service worked. This was key to public efficiency.

Last time…

Now…

Air conditioning was seen as an enabler of productivity and a provider of comfort and quality of life in a tropical climate, before its effects on the environment were fully understood

It is better understood that air conditioning pumps heat into the surroundings and exacerbates the warming that is already happening due to climate change

How can Singapore continue to ensure comfort and productivity whilst not contributing to urban heat?

Historical Context: Shifts In City-wide /National Priorities

- Sometimes past decisions are insufficient for the present/future because priorities shift: what was once seen as not of urgent need becomes valued now e.g. the case of heritage conservation

Last time…

Now…

Built heritage's preservation is seen as crucial to maintaining a uniquely Singaporean identity and sense of belonging

Heritage buildings were torn down to meet the urgent need for housing and spaces for commerce and industry

How can Singapore continue to provide homes and spaces for industry and commerce while preserving more of its built heritage?

Historical Context: Shifts In City-wide /National Priorities

- Changes to national urban planning directions can also raise questions about the sufficiency of infrastructure, designs, policies and programmes that were implemented with a different planning direction in mind e.g. creation of regional centres

Last time…

The Central Area was the main business centre in Singapore

Now…

URA's decentralisation strategy, introduced since the 1991 Concept Plan, has expanded Singapore's business ecosystem with job nodes across the island. This aimed to reduce traffic congestion towards the city centre, gives businesses options, and brings jobs closer to home.

Are Singapore's developing regional centres and gateways equipped with enough amenities, infrastructure and services to support the greater influx of people and businesses that may result?

Historical Context: Shifts In Tastes And Preferences

- Sometimes urban infrastructure, designs, policies and programmes are insufficient because tastes and preferences shift
- E.g. shifts in work patterns

Office buildings were seen as essential for any functioning company

Especially with COVID, work-from-home arrangements have become more acceptable and popular

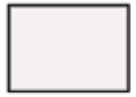
For office-building-intensive areas such as the CBD, there are questions on how to keep these areas relevant at a time when office spaces may not be occupied for long periods of time

Historical Context: Why Look At It?

- Some problems or issues arise out of mismatch between what was planned for and what has happened since
- Comparing the historical intention/purpose with the reality now can help you identify more issues that your site has

Spatial Context

Examining spatial context of a site consists of asking the following:

1. What kinds of land uses are designated by Master Plan?
   - E.g. On the Master Plan, a Dark Blue site (Commercial) vs. a White site allows for different kinds of land use and activities
   - Dark Blue site: only Commercial Commercial
   - White site: signify areas with flexible land use, potentially allowing for commercial, residential, hotel, sports, and recreational developments.  White

Spatial Context

Examining spatial context of a site consists of asking the following:

2. What kinds of people use these designated land uses, and what activities do they engage in there?
   - E.g. Commercial: white-collar workers, office work
   - E.g. White: a whole range of users and activities such as office work, housing, recreation and a whole range of users spanning different social backgrounds

Spatial Context

Examining spatial context of a site consists of asking the following:

3. What do these people and activities need to support them? What amenities do they need? What are the patterns of behaviour that tend to happen due to the nature of the people and activities that use the site?
   - E.g. Commercial: demand for transport at morning/evening peak affecting experience of site; lunch hour causes eateries and nearby shopping areas to be crowded
   - E.g. Residential areas: need schools, playgrounds, supermarkets, clinics, etc. to support the various needs of the resident population

Spatial Context

Examining spatial context of a site consists of asking the following:

4. Find issues: are the needs of the people and activities that take place in these land uses being met? Are there opportunities that arise out of this situation that can be grasped, but are not?

Spatial Context

Note on transportation links:

- Some roads and transport links (walk, cycle, train, bus networks) can enable connectivity between land uses
- However, some features such as highways may also sever connections between land uses
  - E.g. one-north was placed near NUS to create R&D synergies that would span the academic and industry contexts. But the lack of crossings across the AYE has meant that there is a lack of connectivity between the two places

Governance Context

- The governance context can be identified by answering questions such as:
  - Who gets to make decisions and shape this site?
  - What are their aims?
  - Who do these decision makers serve and/or collaborate with?
- Broadly, governance is more than just the government (which includes ministries such as MND and MTI and statutory boards such as HDB and JTC)
- It can include the private sector, NGOs and the public: each of these has varying levels of influence depending on the context

Governance Context

- Consider if certain areas/aspects of the site and its population being neglected due to
  a. A lack of designation of who is in charge,
  b. A lack of clear scope of authority for the people/institutions who are supposed to be in charge,
  c. A neglect by the authorities who are supposed to be in charge and a lack of accountability for this neglect,
  d. Conflicting scopes of authority between different people/institutions,
  e. Or a lack of sensitivity to the needs on the ground by the governing people/institutions?

Governance Context: Why Look At It?

- Some of the potential problems that you may identify in your site may arise due to issues of governance and management
- E.g. famously former PM Lee Hsien Loong mentioned about the incident of a fishball stick in his National Day Rally 2014:

"One resident saw a fishball stick on the walkway one day. The next day, the same stick was still there. So Yen Ling (referring to Mayor Low Yen Ling) called up the agencies, to find out why the area was not being cleaned regularly... Here is what she found: (1) On the left of the walkway is a slope, under NEA; (2) In the middle is a park connector, under NParks; (3) On the right is a pavement next to the road, under LTA; (4) The different cleaners each had their own cleaning schedules; (5) The fishball stick was on the right side, which was only cleaned every two days"

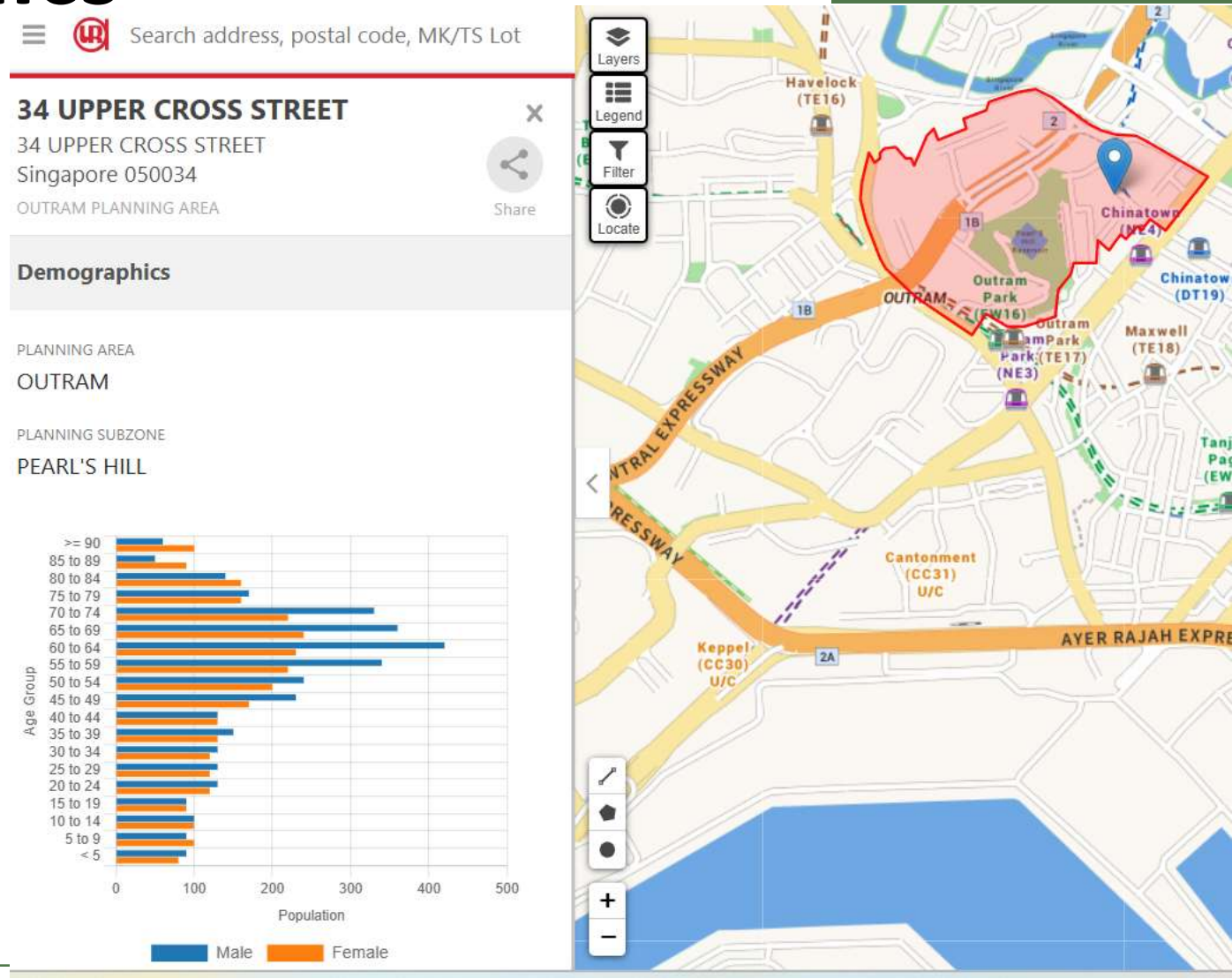
Site context: Summary

- Site context is related to finding potential issues in your site in the following ways:
  1. Socio-economic context: Are the needs of particular social groups being met?
  2. Historical context: Has your site adapted to new conditions?
  3. Spatial context: Are the needs of particular types of land use being met?
  4. Governance context: Are certain areas/aspects of the site and its population being neglected due to the way the site is being managed?
     a. A lack of designation of who is in charge,
     b. A lack of clear scope of authority for the people/institutions who are supposed to be in charge,
     c. A neglect by the authority

ASSIST
Dr. Josh Mini Workshop

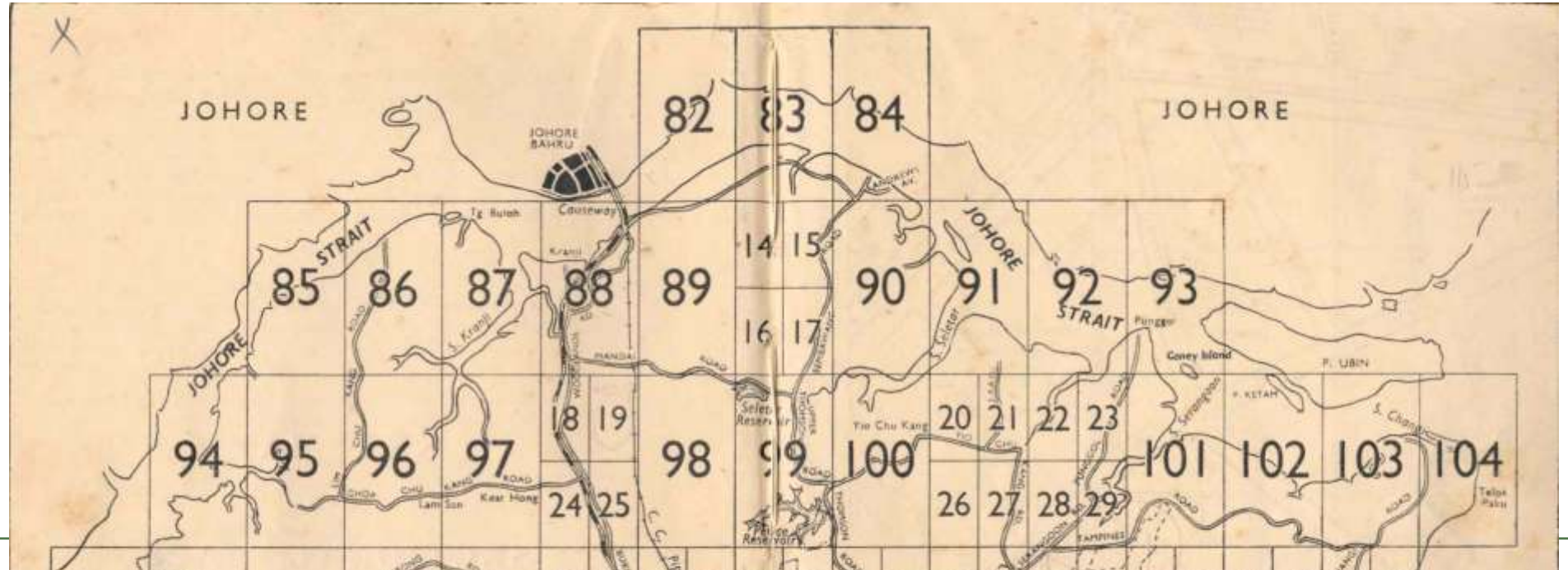
Tools for demographics

- URA SPACE
- Click on the 3 bar menu icon. Scroll down and click on Demographic. Select the area you wish to investigate.



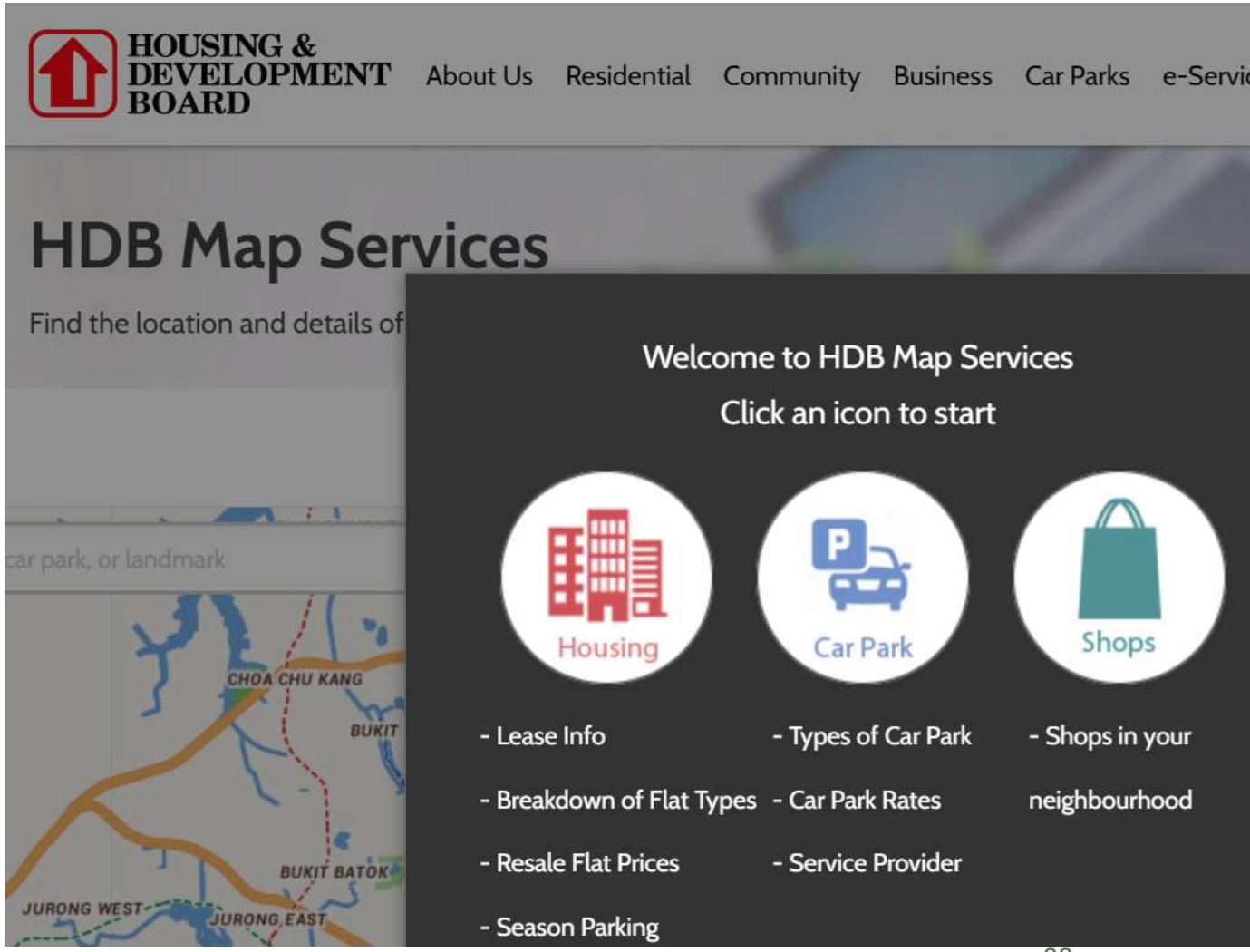
Tools for history

- SLA Maps - https://www.onemap.gov.sg/historicalmaps/
- Postal district
- Infopedia



Tools for demographics

- HDB Map Services
- Other tools? – Google map, Onemap, GenAI



Problem Statements

What is a Problem Statement?

- A statement that clearly and concisely defines the issue within your site
  - A good problem statement should highlight the importance of your issue and summarise key aspects that your audience can pay attention to
  - *It should include the causes of your issue & the people it would affect*
- Helps direct the project towards finding a solution that would address the problem and involve relevant stakeholders
  - Connects the current state of your site to the goal or desired outcome of your site

This problem statement will form the foundation of your project and subsequent presentations

Developing a Problem Statement

1. Think: what would a **high quality of life**, **competitive economy** and a **sustainable environment** (the Liveability outcomes) at your site looks like (not just to you, but to those who live, work and play there)?

2. Through your **preliminary site visit and analysis**, understand the current state of the district and **identify problems** within– how does your site miss the mark of the Liveability outcomes?

3. **Ask why:** why is this problem there? **Are there deeper problems** that underlie and which cause the problems that you have identified?

4. Craft a **specific** problem statement that addresses the **deeper, more underlying problems** (Why?) and addresses the **specific context** (Who? Where? When? What?) of the problem

Example

There is a lack of **vibrancy** in the Downtown core.

This implies that the Downtown core lacks vibrancy.
*But what **specific** kind of vibrancy does it lack?*

There is a lack of **creative artistic activity** in the **Downtown core**.

*Are there **specific situations** in which you have observed that a lack of creative artistic activity is a concern? For certain **people**? For certain **times**? At certain **places**? For certain **reasons**?*

There is a lack of **creative artistic activity** for **youth** in the **areas of the Downtown core** where **office-use and shopping currently predominate.**

All in one idea

Context – Social, economic, spatial patterns

Urban systems – Various key systems involved.

Story – Make a data backed and believable pitch

Ideal – Ideal state for who and why?

Project – Ideal and feasible state for who and why in the near future.

Baseline – Existing policies/ initiatives/ situations

Useful resources on research methods

Creating Surveys

- Before creating a survey, consider the following:
  - **Who** do you want to target with this survey? (e.g. shop merchants in Bedok Town Centre/ residents in Tengah)
  - **What** do you want to learn from this survey? (e.g. shop merchants and their opinions on e-commerce? Tengah residents and the issues faced in an estate still under development?)
    - Additional consideration: Are you trying to understand severity/ extent of an issue? Are you trying to see what is currently available to address this issue?

Designing Your Survey

- Keep your survey **short (5-10 questions)** to maximise response & completion rate
- **Use neutral language**
  - Avoid **leading questions** – these questions denote your personal opinions and are likely to induce a biased response
  - For eg: "Why is the new bus route better than the previous one?"
    - This may cause respondents to question whether the creation of the new bus route was to address problems associated with the previous bus route (even though this may not be the case)
    - You can consider rephrasing this question to "how do you feel about the changes made to the bus route?"

Designing Your Survey

- **Be as clear and specific as possible**
  - Don’t ask double-barrelled questions – these questions ask for more than one topic but only allow for one response.
  - For eg "What are your thoughts about the bus and train routes?"
    - For this question, respondents are prompted to share one response to cover their thoughts about both the bus and train routes (even though they may have different thoughts about each item mentioned).
    - You can reconsider breaking this question down into two separate questions

- Ask questions in logical order – **funnel approach**
  - Put broadest, more general questions at beginning
  - Go into more specific questions in middle
  - Closed questions such as asking a person’s age can go towards end

Distributing Your Survey

- **Choose a distribution method:** In-person at site or through online platforms (e.g. Google Forms, SurveyMonkey, Qualtrics)
- **Targeted outreach:** Reach out to specific groups relevant to your research objectives. One way to start is snowball sampling, using your current social networks.
- **Additional considerations while distributing your survey:**
  - Inform respondents about purpose of survey, how their data will be used, and their right to withdraw
  - Anonymity and confidentiality: Ensure respondents that their responses are anonymous and confidential
  - Sensitive questions: Handle sensitive questions with care, providing options for respondents to skip questions if they feel uncomfortable

Preparing for Interviews

- **Purpose:** understand public perceptions of district and/or topic, to complement findings from desktop research

- **How to prepare for interviews:**
  - Clearly outline goals of interview and information you are seeking
  - It is fine to have broader objectives in initial stages of formulating project
    - Questions like “What do I want to take away from this interview?” and “How will responses from the interview inform my understanding of my site?” help with crafting interview questions

- **Develop your interview questions:**
  - Create list of open-ended questions tailored to intended respondents
  - When interviewing different groups (e.g. bus drivers and commuters), prepare separate questions

Developing Your Interview Questions

- **Craft topic-specific/focused questions:** Instead of asking “Tell me about the district”, ask “What are some areas for improvement for the public transport infrastructure of the district?”

- **Use the funnel approach** to structure your interview: broad questions at the beginning, more specific questions in the middle and closed questions towards the end .

- **Avoid asking leading questions:** Instead of asking “Why is the new bus route better than the previous one?”, you may rephrase it to “How do you feel about the changes made to the bus route?”

- **Encourage detailed responses:** Add in prompts into your question. If you are looking at public transport infrastructure, instead of just asking about the areas for improvement, ask/follow-up with a “why”, or “what do you mean by ...?”.

- **Be neutral and non-judgmental:** Instead of asking “Why don’t you use the public transportation system?”, ask “How would you describe your experience with using or choosing not to use public transportation?”

During Your Interview

- **Considering interviewing people in pairs** – one person can engage with the interviewees while the other person can take notes (key points, quotes and observations during the interview)
  - These roles do not need to be fixed! Feel free to change up who's doing what anytime.

- **During an interview**
  - **Build rapport with your interviewees:** Ask them about their backgrounds if relevant to project e.g. age, purpose of being in the area
  - **Be mindful of your body language** during your interview : ensure that your body is on the same level as the person you're interviewing, maintain constant eye contact
  - **Use your prepared questions as guide,** but be flexible to follow interesting leads
    - Interviewees may bring up themes or concepts that your group wouldn't have thought of before!
  - **Pay close attention** to what your interviewees have to say, and ask relevant follow-up questions

- Once interview has concluded, **thank respondents for their time**

Surveys vs Interviews

- Similar ways to prepare for both research methods
- Major differences are in ways of collecting information, type of data collected, and analysis of data
- Method chosen depends on nature of project

	Survey	**Interview**
Research Objectives	Quantitative insights required	Qualitative insights are required
Sample Size	Large sample sizes, broader data collection across bigger and diverse groups	Smaller and targeted sample sizes / depth of understanding is crucial
Type of Information Needed	Numerical measurement of attitudes, behaviours or preferences	Explore personal experiences and opinions
Resource & Time Constraints	More cost-effective and efficient for large scale data collection	More time and effort required per respondent

References

- Parfitt, J. (2005). Questionnaire design and sampling. In R. Flowerdew & D. M. Martin (Eds.), Methods in human geography (pp. 78-109). Routledge.
- Tunison, S. (2023). Observational study. In J. M. Okoko, S. Tunison, & K. D. Walker (Eds.), Varieties of qualitative research methods (pp. 45-67). Springer.

Reminders

Next Friday (6 Feb): Seminar 3

- Topic: **Sustainable Environment + Systems Thinking and the Causal Loop Diagram**
- Lecturers: **Dr. Rosita Samsudin** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Lecturer, NUS Cities; PhD, Architecture, NUS); **Dr. Corinne Ong** (Senior Lecturer, Residential College 4 (RC4))
- This seminar will go into more depth on what a **Sustainable Environment** consists of. This is crucial to understand in your projects as you will be assessing how liveable your site is. Understanding the gap between the ideal of a Sustainable Environment and the current state of your site can help you to start identifying potential issues that you can work on in your project.
- This seminar will also cover what **systems thinking** is and introduce the **Causal Loop Diagram (CLD)** as a tool to uncover the underlying systems and causes behind an urban problem so that you can apply it to the Problem Statement you will identify.
- **IMPORTANT TO NOTE: For students who have been allocated in-person attendance for Seminar 3, please make sure to be presentbecause you will have to do the quiz during the seminar (this quiz will not be made visible to those attending online) - quiz will be held DURING seminar**

Next Friday (6 Feb): Quiz 2

- **IMPORTANT TO NOTE: For students who have been allocated in-person attendance for Seminar 3, please make sure to be present because you will have to do the quiz during the seminar (this quiz will not be made visible to those attending online) - quiz will be held DURING seminar**
- **Only those allocated for in-person attendance to Seminar 3 will have to do Quiz 2**
- **If you have not been allocated for in-person attendance to Seminar 3, you will not have to do Quiz 2**
- **Topic of Quiz:** Seminar content of Seminar 3 (so pay attention to the seminar!)
- **Quiz Format:** On Canvas, 8 minutes, 5 MCQs

Next Friday (6 Feb): Individual Assignment 1 Opens

- **Format:** On Canvas, One short-answer question, 5 marks
- **Topic:** Reflection on Seminars 1 to 3
- **Where to find:** Canvas coursepage > Sidebar > Assignments > Individual Assignment 1
- **Weightage:** will contribute 6.25% to your final grade
- Assignment will be open from **Friday 6 Feb 12:00pm to Friday 13 Feb 11:59pm** (Please make sure to complete by this time!)
- **PLEASE NOTE:** You are only allowed one attempt, so please plan your answer first before starting the assignment and submitting on Canvas. However, there is no time limit for the attempt. The attempt only ends when you submit your answer on Canvas. Please also **click the 'Submit' button! Don't let your efforts go to waste.**

Next Studio: Studio 2 (Consultation) [Week 5]

Deliverables to Bring to Studio:

- Please prepare any photos, observations, notes that you have gotten so far
  - Note: you should be doing your first site visit before next studio! (refer to the Site Visit slides for guidance)
- We will be engaging in class activities to consolidate your observations to enable you to have a deeper understanding of the site issues

Next Studio: Studio 2 (Consultation) [Week 5]

Studio Outcomes:

- Understand the needs, behaviours, goals and challenges of key stakeholders within your allocated site
- Compare the current state of the site (as observed and researched) to the ideals of the Liveability Outcomes of the Liveability Framework (High Quality of Life, Sustainable Environment, Competitive Economy)
- Apply causal loop diagrams to map and interpret the interconnections, feedback mechanisms and dynamic relationships among urban systems
- Craft a problem statement that reflects (an) issue(s) in your site that your team wishes to tackle for the rest of the project.